When Does Intraday Options Trading Predict Stock Returns?*

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August 28, 2026

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Keywords: Options order imbalance; intraday trading; stock return predictability; options informed trading; liquidity timing
JEL Classification: G12; G13; G14

* An earlier version was titled "Intraday Signed Options Volume and Stock Return Predictability." We are grateful to Kalok Chan, Hui Chen, Tarun Chordia, Kewei Hou, Dmitriy Muravyev, Jun Pan, Neil Pearson, Avanidhar Subrahmanyam, Fred Sun, Lai Xu, and Chen Yao, as well as conference and seminar participants at China Derivatives Youth Forum and The Chinese University of Hong Kong for constructive comments. We thank CBOE LiveVol and Ellen Spiro for providing the CBOE open/close data and the International Securities Exchange and Conrad Chow for providing the ISE open/close data. Any remaining errors are our own.
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Abstract

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1. Introduction

How and why do option trades predict underlying stock returns? A stream of literature focuses on options volume predicting subsequent underlying stock returns, including put-call ratio (Pan and Poteshman, 2006; Weinbaum et al., 2023), option-to-stock (O/S) trading volume (Johnson and So, 2012; Ge, Lin, and Pearson, 2016; Neuhierl et al., 2026), and options order imbalance (OOI) measures (Hu, 2014; Bryzgalova, Pavlova, and Sikorskaya, 2023; Cao et al., 2026).[1] Although these weekly or daily option volume-based measures capture the excess demand of informed options traders, they are too coarse to identify precisely when the information is revealed and incorporated into stock prices. An intraday analysis can thus shed light on the information-revelation process by tracing the timing of signed options volume, separating anticipation from execution, and determining whether return predictability emerges within the day. Yet such intraday patterns of signed options volume in relation to stock returns predictability remain underexplored.[2]

Our paper extends this line of research by combining high-frequency signed options volume from the Chicago Board Options Exchange (CBOE) and the International Securities Exchange (ISE) to construct intraday OOI measures and examine their cross-sectional predictive power for future stock returns.[3] The gist of the empirical design is that informed traders face a trade-off between information advantage and price impact (Holden and Subrahmanyam, 1994). For example, the concentration of uninformed trading near the market close enables them to trade with reduced price impact, albeit at the cost of some informational advantage. Hence, the return predictability

[1] Another stream of literature uses option price-based measures, including volatility spread (Bali and Hovakimian, 2009), deviations from put-call parity (Cremers and Weinbaum, 2010), implied volatility smirk (Xing, Zhang, and Zhao, 2010), changes in implied volatilities (An et al., 2014), and risk-neutral skewness (Chordia, Lin, and Xiang, 2021). The negative side against price-based measures includes Chan, Chung, and Fong (2002), Muravyev, Pearson, and Broussard (2013), Goncalves-Pinto et al. (2020), Ni et al. (2021), and Muravyev, Pearson, and Pollet (2025a).

[2] We do acknowledge that Hu (2014) conducts an intraday analysis of OOI but excludes the trading near the market open and close. Bergsma et al. (2020) find the first 30-min options trading predicts the rest-of-day stock returns but remain silent about remaining intraday intervals, long-term stock returns predictability, and liquidity timing.

[3] During our sample period, CBOE and ISE account for 31% and 23% of total options trading volume, respectively.

of intraday OOI may vary within a day. Empirically, we examine next-day stock return predictability of 30-minute OOI during regular trading hours (9:30 a.m. to 4:00 p.m.). We find that OOI exhibits statistically significant predictive power across all intervals. On average, a one-standard-deviation increase in OOI predicts a 1.695 basis points (bps) increase in next-day stock returns (4.36% per annum). More importantly, this predictive power intensifies near the market close: a one-standard-deviation increase in OOI during the last half hour predicts a 3.543 bps increase in next-day stock returns (9.34% per annum), more than twice the average magnitude.

We then decompose the overall OOI into several subcomponents. First, Johnson and So (2012) argue that synthetic short positions are more informative because option markets allow informed traders to circumvent short-sale constraints in the equity market. In contrast, Ge, Lin, and Pearson (2016) find that synthetic long positions contain comparable or even greater informational content than short positions. Consistent with Ge, Lin, and Pearson (2016), we find that the predictive power of call OOI is comparable to that of put OOI. Second, Pan and Poteshman (2006) argue that trades opening new positions are more informative than those unwinding existing positions. We find corroborative evidence that OOI constructed by option trades opening new positions has stronger predictive power than OOI constructed by option trades closing current positions.

We further investigate how the documented predictability varies with option moneyness. On the one hand, if options with higher implicit leverage tend to attract more informed traders, such options should have stronger return predictability (Black, 1975). Cao, Chen, and Griffin (2005), Pan and Poteshman (2006), and Ge, Lin, and Pearson (2016) show that the predictive power of signed options volume is indeed more pronounced for high-leverage options, i.e., out-of-the-money (OTM) options. On the other hand, Johnson and So (2012) and Hu (2014) demonstrate that the predictability is stronger for low-leverage options, i.e., at-the-money (ATM) and in-the-money

(ITM) options, plausibly due to their higher liquidity. Adding to this trade-off between leverage and liquidity, we find that the return predictability mainly comes from OTM options, especially near the market close, which squares with the leverage story. Nevertheless, ATM and ITM OOI also exhibit predictive power, though less persistently, suggesting that some informed traders may prefer these contracts to avoid the higher transaction costs associated with OTM options.

Besides signed options volume, a key feature of the CBOE and ISE Open/Close data is the categorization of trading volume by investor type. Exploiting this data feature, we construct the OOI using options trades from "small customers," "middle customers," "large customers," and "firm."[4] We find that the predictive power of OOI for stock returns is almost only significant for small customers (i.e., small-size orders). This result is in line with the order splitting of informed traders (Kyle, 1985; Easley and O'Hara, 1987) or the emergence of informed retail options investors (Eaton et al., 2026).

To better understand the underlying economic mechanisms, we follow Hu et al. (2025) and aggregate the original thirteen 30-minute OOI intervals into three broader intraday segments, including (i) the opening period (the first four half-hour intervals), (ii) the midday period (the middle five intervals), and (iii) the closing period (the final four intervals). This coarser temporal resolution facilitates more interpretable analysis of order imbalance dynamics across trading day phases. Consistent with our baseline results, the closing period OOI exhibits the strongest predictive power. It also generates the highest out-of-sample R-squared and better forecast performance compared to other intervals. We further document that the intraday pattern in the predictive power of OOI becomes particularly pronounced after the onset of COVID pandemic and persists when forecasting returns over longer horizons to at least four weeks.

[4] "Small", "middle", and "large customers" categories refer to option trades with volume less than 100, 100-199, and greater than 199 contracts each. "Firms" category combines option trades from clearing firms of options clearing corporation (OCC) and broker-dealers. See Section 2.1 for definitions of these investor types.

An important follow-up question is why intraday OOI predicts future stock returns. One explanation is that options trading conveys private information flow from option market to stock market. Alternatively, other types of informed traders may simultaneously incorporate information into stock prices while uninformed investors drive trading volume in the options market. We begin by examining the relationship between options trading and other well-documented forms of informed trading. If signed options volume contains no unique information and merely reflects the presence of other informed investors, e.g., institutions, short-sellers, and insiders, the predictive power of OOI should diminish once we control these alternative trading activities. However, we find none of other informed traders fully subsumes the predictive power of OOI.

In particular, the predictive power of OOI near the market close is stronger for stocks with higher transient ownership and weaker for those with higher quasi-indexers ownership.[5] It suggests that option-informed trading is positively correlated with active institutional trading. In addition, we find that OOI near the close more strongly predicts returns for stocks facing greater short-sale constraints and higher borrowing costs. In contrast, informed options traders do not overlap insider transactions. These results suggest that while informed investors may operate in both options and equity markets, options trading still conveys distinct information about future stock returns.

After excluding this alternative mechanism, we turn to examine the informed trading channel driving the stock return predictability of OOI. We first examine whether informed option traders exhibit the liquidity timing behavior proposed by Collin-Dufresne and Fos (2016) and Kacperczyk and Pagnotta (2019). Specifically, we find that OOI's predictive power near the close is stronger on equity option expiration days on the third Friday of each month, and on "Triple Witching" days when stock market index futures and options, and stock options all expire. These days are typically

[5] Transient and dedicated institutions are particularly active in short- and long-term trading, respectively, while quasi-indexers are more likely to passively trade to minimize tracking errors. More details are in Bushee (1998).

associated with higher equity options trading volume near the close, driven by the rollover of expiring p.m. settled contracts.[6] Although this expiration-induced trading contains no information, it provides a noisy trading environment for informed options traders to camouflage their information. Moreover, we find that the predictive power of OOI is consistently stronger when options liquidity is elevated, but weaker when underlying stock liquidity is higher. This joint liquidity timing pattern is consistent with the implications in Easley, O'Hara, and Srinivas (1998).[7]

Within the informed trading channel, an important question is what types of information are incorporated through options trades. Prior studies document that the predictive power of options trading is stronger on days with firm-specific news, implying the incorporation of new information into stock prices. Notable examples include M&As (Chan, Ge, and Lin, 2015), analyst events (Lin and Lu, 2015), earnings announcements (Bergsma et al., 2020), and broader corporate disclosures (Chordia, Lin, and Xiang, 2021; Cheng et al., 2026). In addition, Jin, Livnat, and Zhang (2012) argue that options traders possess a superior ability to process public information compared to equity traders. Building on these studies, we examine how the predictive power of OOI varies in response to firm-specific information, distinguishing between private and processed public information channels. We classify a stock-day as containing private information if firm-specific earnings or analyst-related unscheduled news will be released during the overnight following the trading day, and as containing processed public information if such news was released during the overnight preceding the trading day. We show that informed options traders respond to prior public

[6] Another important driver is Robinhood's forced liquidations (Amaya et al., 2026). Robinhood will automatically liquidate ITM option positions from 15:00 to 15:05 (15:30 to 15:35 after December 9, 2022) when customers lack the cash or shares required to exercise contracts at expiration. The uninformed forced liquidation may also explain why the predictive power of OOI near the close is higher after COVID-19, as retail options trading skyrockets afterwards.
[7] Easley, O'Hara, and Srinivas (1998) argue that when options liquidity is low, the profit to informed traders trading in stock market is higher, so option market functions only to handle liquidity-based trading, resulting in a "separating equilibrium." In contrast, when options liquidity is high and/or the stock liquidity is low, informed traders will use options and reach a "pooling equilibrium" in both markets.

information by trading immediately after the market open, while exploiting the noisy and liquid environment near the close to trade on private information about forthcoming news. Our results shed new light on how options traders respond to different information channels within a day.

Beyond firm-level events, we further examine the intersection between options trading and key macroeconomic events, which remains an open question. We extend the baseline model by interacting OOI with indicators for Federal Open Market Committee (FOMC) meeting days, as well as daily absolute changes in one-year zero-coupon bond yield and VIX index. Our findings suggest that informed options traders respond not only to public macro news, such as FOMC announcements and press conferences, but also to private macro information not yet publicly disclosed. Specifically, the predictive power of OOI is elevated on the day before FOMC meetings and on days characterized by heightened uncertainty and interest rate risk, highlighting the options market's broader role in processing both public and private macroeconomic signals.

We conclude our empirical analysis with a set of additional tests designed to reinforce and extend our main findings. First, following Pan and Poteshman (2006), we examine the predictive power of open-buy put-call ratio (OBPC) for next-day stock returns and find that OBPC near the market close exhibits the strongest forecast power. Second, we find the OOI near the market close can positively forecast next-day stock price jumps and lower future realized skewness, suggesting that signed options trading volume near the market close also contains information about downside tail risk. Third, the intraday pattern in the predictive power of OOI remains robust after controlling for the stock borrow fee,[8] under alternative specifications with two-way fixed effects in a panel regression framework, and when OOI is constructed solely from data on either the CBOE or ISE.

Our contributions to prior literature are twofold. First, to the best of our knowledge, we are

[8] Stock borrow fee is unlikely to explain the intraday variation of OOI's predictive power. First, the public disclosure of stock borrow fee is typically at low frequency, e.g., daily (such as Markit). Second, Muravyev, Pearson, and Pollet (2025b) show that the borrow fee for a stock is highly persistent in time-series with a 97% autocorrelation.

the first to document how the stock return predictability of options trading varies throughout the full intraday period. Previous studies either operate at a lower frequency or omit the market open and/or close hours.[9] For example, Hu (2014) excludes one hour around the market open and close to avoid estimation errors of Lee and Ready (1991) algorithm. Bergsma et al. (2020) focus only on the first half-hour O/S in predicting rest-of-day returns. Easley, O'Hara, and Srinivas (1998) treat the predictive power of intraday signed options volume as uniform across intervals. Our results differ from these studies, as we clearly depict an intraday variation in the predictive power of signed options volume, highlighting the role of informed options trades around the market close.

Second, we also contribute to the literature that examines the information content of options trading around corporate events (Cao, Chen, and Griffin, 2005; Jin, Livnat, and Zhang, 2012; Chan, Ge, and Lin, 2015; Lin and Lu, 2015; Chordia, Lin, and Xiang, 2021; Liu, Peng, and Tang, 2023; Cheng et al., 2026). We extend this work by separately examining private and processed public information channels around firm-specific news, documenting intraday heterogeneity in when informed options traders incorporate different information channels. Furthermore, existing evidence on the role of macro news in informed options trading remains limited. We extend this literature by documenting that the stock return predictability of options trading is stronger around macroeconomic shocks. We also contribute to Easley, O'Hara, and Srinivas (1998), Collin-Dufresne and Fos (2016), and Kacperczyk and Pagnotta (2019) by showing that informed options traders appear to time liquidity in both option and underlying stock market under an intraday setup.

The rest of this paper is organized as follows. Section 2 details data selection, variable constructions, and specifications. Section 3 illustrates the stock return predictability patterns of

[9] Low-frequency option volume-based literature includes Pan and Poteshman (2006), Roll, Schwartz, and Subrahmanyam (2010), Johnson and So (2012), Ge, Lin, and Pearson (2016), Chen, Joslin, and Ni (2019), and Weinbaum et al. (2023); option price-based literature includes Bali and Hovakimian (2009), Cremers and Weinbaum (2010), Xing, Zhang, and Zhao (2010), Stilger, Kostakis, and Poon (2017), and Chordia, Lin, and Xiang (2021).

intraday OOI. Section 4 shows evidence of the informed trading channel behind the predictive power of OOI. Section 5 conducts additional tests to extend our results. Section 6 concludes.

2. Data, Variables, and Specifications

2.1. Data and Sample

We obtain signed options trading volume from the CBOE and ISE Open/Close Trade Profile Intraday, which provides 10-minute data on both quantity and volume for each individual option. Both exchanges clarify quantity and volume into four types: open buy (open sell) indicates that investors are buying (selling) options to open new long (short) positions, while close buy (close sell) indicates that investors are buying (selling) options to close current short (long) positions. Each option trade is further tagged with different investor types.[10] Since all investors can engage in informed trading, we aggregate the trading volume of all investor types except market makers (Bondarenko and Muravyev, 2025).[11] We also aggregate the trading volume across two exchanges for each individual option to obtain a more comprehensive sample.

We collect other data from a variety of databases. Intraday transaction data of individual stocks comes from the NYSE daily Trade and Quote (TAQ). We merge TAQ and the Center for Research in Security Prices (CRSP) to obtain the unique stock identity (PERMNO) and adjust the TAQ prices for stock splits and stock dividends. We collect daily options trading information from

[10] CBOE Open/Close database includes five investor types: (1) customers; (2) professional customers; (3) firms; (4) broker/dealer; and (5) market maker. ISE Open/Close database includes three investor types: (1) customers; (2) professional customers; (3) firms (which is further split into deeper types, i.e., priority and broker/dealer). A professional customer is a special type of customer: anyone who is not trading as a market maker or firm account that enters more than 390 orders per day over the course of one month will be assigned as a professional customer. The first two customer types are further split into "small" (<100), "medium" (100-199), and "large" (>199) groups based on the number of contracts each. The market makers have no open and close direction, but always trade against other investors, thus not included in our analysis. We combine the non-professional and professional customers since the latter's trading volume is almost negligible compared to the former. We combine the firms (priority) and broker/dealer into the "firm" category. Detailed definitions can be found on Online Appendix A, CBOE and ISE official websites.

[11] In Section 3.4, we repeat our baseline specification conditional on different investor types.

OptionMetrics, institutional investor holdings from the LSEG (13F) Holdings dataset, short-sale trading from S&P Global Markit, and insider options transactions from LSEG Insider Form 4. Finally, RavenPack 1.0 provides data on public released firm-specific news.

The sample consists of all common stocks (CRSP share codes 10 or 11) from July 2017 to December 2022. We exclude stocks with closing prices lower than $1 and trading days with early stock markets close at 1:00 p.m. We exclude stock-days with zero options trading. Following Hu (2014), we exclude options that expire within ten days and the following trades: (i) off-hour trades; (ii) data errors such as trades with non-positive strikes. After applying these filters, the final sample contains over 1,900,000 stock-day observations spanning 3,317 stocks and 1,374 trading days.

2.2. Construction of Options Order Imbalance

The main independent variable is options order imbalance (OOI), defined as (B−S) / (B+S), where B (S) denotes aggregate buyer-initiated (seller-initiated) trading volume (Easley, O'Hara, and Srinivas, 1998; Hu, 2014; Bernile, Hu, and Tang, 2016). We construct the following bullish OOI to capture informed trading in the option market:

$$OOI_{i,d,t}=\frac{\left(VOL_{i,d,t}^{C,B}+VOL_{i,d,t}^{P,S}\right)-\left(VOL_{i,d,t}^{C,S}+VOL_{i,d,t}^{P,B}\right)}{\left(VOL_{i,d,t}^{C,B}+VOL_{i,d,t}^{P,S}\right)+\left(VOL_{i,d,t}^{C,S}+VOL_{i,d,t}^{P,B}\right)} \quad (1)$$

where $VOL^{C,B}$ ($VOL^{C,S}$) denotes the sum of open-buy and close-buy (open-sell and close-sell) call options volume with underlying stock i, and in time interval t of day d. $VOL^{P,B}$ ($VOL^{P,S}$) denotes the sum of open-buy and close-buy (open-sell and close-sell) put options volume with underlying stock i, and in time interval t of day d. In the baseline regression, we measure OOI at the 30-minute frequency. For example, t = 10:00 corresponds to the 30-minute level from 9:30 a.m. to 10:00 a.m. OOI reflects a bullish view on underlying stock returns. A positive (negative) OOI indicates relatively higher (lower) volume of longing call options than longing put options. An OOI of zero

implies market neutrality, i.e., the same volume of longing call and put options. A special case of market neutrality is zero options trading volume. However, since the denominator of OOI is zero in this scenario, OOI is set as missing and will be omitted from the regression.

2.3. Research Design

In the baseline model, we use each 30-minute OOI computed between 9:30 a.m. and 4:00 p.m. to predict next-day stock returns. In subsequent sections, we aggregate intraday options trading into three coarser intervals, near the market opening (9:30 a.m. to 11:30 a.m.), midday (11:30 a.m. to 2:00 p.m.), and near the market closing (2:00 p.m. to 4:00 p.m.), and re-calculate the corresponding OOI. To avoid the bid-ask bounce, stock returns are calculated using National Best Bids and Offers (NBBO) quote midpoints.[12] The cross-sectional regression (2) is specified as follows:

$$Ret_{i,d+1} = \alpha + \beta \times OOI_{i,d,t} + \psi' X_{i,d,t} + \varepsilon_{i,d+1} \quad (2)$$

where $X_{i,d,t}$ is a battery of stock- and option-level control variables that prior literature has shown to predict stock returns: stock order imbalance ($SOI_{i,d,t}$), which captures informed trading in the equity market (Chordia and Subrahmanyam, 2004); the natural logarithm of stocks ($StockVol_{i,d,t}$) and options trading volume ($OptionVol_{i,d,t}$), the stock bid-ask spreads ($Spread_{i,d,t}$), and the monthly average of daily stock turnover at the most recent month end ($Turnover_{i,d}$), motivated by Hu (2014) to control for liquidity. We also include annualized volatility at the most recent month end ($Volatility_{i,d}$), the natural logarithm of market capitalization ($Size_{i,d}$) and book-to-market ratio ($BTM_{i,d}$) at the most recent quarter end. Stock returns yesterday ($Ret_{i,d-1}$), over day $d-20$ to $d-2$ ($Ret_{i,d-20,d-2}$), and over day $d-120$ to $d-21$ ($Ret_{i,d-120,d-21}$) control for short-term reversal and long-

[12] Our results remain robust if we calculate stock returns using closing prices. We use next-day stock returns to keep the return prediction task constant and to directly compare the predictive power of OOI over different intervals. We also consider predicting the rest-of-day stock returns from t to 4:00 p.m. and the next half-hour stock returns from t to $t+1$. Tables OA2 and OA3 in Online Appendix report the results, respectively.

term momentum, respectively. Detailed variable definitions are provided in Online Appendix A. Because OOI captures bullish sentiment, we expect $\beta > 0$: an increase in OOI, i.e., relative increase in longing call options and/or shorting put options, should predict higher subsequent stock returns.

Panel A of Table 1 presents summary statistics of selected stock- and option-level variables. The average daily stock return is 2.176 bps The average OOI (SOI) across three intervals is −0.020 (−0.020), −0.025 (−0.011), and −0.008 (−0.005), respectively. Panel B shows the correlation between order imbalances. The correlation and autocorrelation of OOI (SOI) across three intervals are positive. Interestingly, OOI and SOI are negatively correlated with a correlation of −0.0306. Hu (2014) states that an underestimation of structured trades in the sample, e.g., the covered calls and the protected puts, could account for this negative correlation. Figure 1 shows the average options trading volume of CBOE and ISE for each 30-minute intraday interval. The options trading volume has a U-shape and comes mainly from small customers.

3. Stock Returns Predictability of Intraday OOI

3.1. The 30-minute OOI

We estimate Fama and MacBeth (1973) cross-sectional regression (2) to examine the predictive power of 30-minute OOI for next-day stock returns. Table 2 reports the average slope coefficients and corresponding Newey and West (1987) *t*-statistics with 6 lags.[13] From column (1) to column (13), the coefficient of OOI is significantly positive in all thirteen intervals, indicating that higher OOI predicts higher next-day stock returns. To quantify the average economic magnitude, we stack thirteen 30-minute intervals together, and report the results in column (14), which yields a significantly positive coefficient of 1.695. The magnitude is economically essential: a one-

[13] Following Greene (2011), we set the optimal lag as the integer part of $T^{1/4}$, where T is the number of trading days in our sample period. Our results are robust with alternative lags, such as the integer part of $0.75*T^{1/3}$.

standard-deviation increase in 30-minute OOI predicts, on average, a 1.695 bps (*t*-stat. = 15.12) increase in next-day stock returns (4.36% per annum). Importantly, in column (13), a one-standard-deviation increase in the *last* half-hour OOI on average predicts a 3.543 bps (*t*-stat. = 9.73) increase in next-day stock returns (9.34% per annum), which is 114% larger than the average effect.[14]

To visualize the comparison, Figure 2 plots the standardized OOI coefficients, robust *t*-statistics, and corresponding 95% confidence intervals across thirteen intraday intervals. The last OOI's coefficient is significantly larger than the average OOI coefficient and than each of the remaining twelve coefficients at the 5% level. These findings demonstrate a persistent predictive power of intraday OOI for next-day stock returns and highlight the importance of options trading near the market close. These findings contribute to both the low-frequency options literature and the high-frequency analysis by Hu (2014) and Bergsma et al. (2020).[15]

Turning to the control variables, the coefficient of SOI in column (13) is significantly positive at 1.057 (*t*-stat. = 2.65) but is weaker in both statistical significance and economic magnitude compared to the coefficient of OOI. Stocks with higher stock or options trading volume tend to earn lower subsequent returns due to the prompt detection and elimination of potential mispricing (Brennan, Chordia, and Subrahmanyam, 1998). Lagged stock returns exhibit a short-term reversal and long-term momentum effects, consistent with Jegadeesh (1990). In subsequent tables, we omit the reports of constants and control variables for brevity.

3.2. Decompositions of OOI

An intrinsic advantage of options is that both call and put options offer embedded leverage, while put options additionally allow traders to bypass short-selling constraints in the underlying stock

[14] Table OA4 horse races OOIs and shows that the last OOI's predictive power is not subsumed by remaining OOIs.
[15] Our results do not contradict Bergsma et al. (2020), who find O/S near the market open has the strongest predictive power for rest-of-day stock returns. As shown in Figure OA2 and Table OA2, OOI near the market open does have stronger predictive power for rest-of-day stock returns compared to remaining intervals. Instead, our paper focuses on next-day stock return predictability and option trading near the market close.

market. We first respectively construct OOIs using call and put options only and horse race their predictive power to shed light on the debate on the information component between call and put options trading. The decomposition in each 30-minute interval follows equations (3):

$$OOI_Call_{i,d,t}=\frac{VOL_{i,d,t}^{C,B}-VOL_{i,d,t}^{C,S}}{VOL_{i,d,t}^{C,B}+VOL_{i,d,t}^{C,S}},\ OOI_Put_{i,d,t}=\frac{VOL_{i,d,t}^{P,B}-VOL_{i,d,t}^{P,S}}{VOL_{i,d,t}^{P,B}+VOL_{i,d,t}^{P,S}} \tag{3}$$

Parallel to total OOI, call OOI is a bullish measure. Therefore, we expect a positive predictive power. Put OOI, by contrast, is a bearish indicator, and we expect it to negatively predict stock returns. More importantly, a horse-racing setup can distinguish the return predictability between call and put OOI. We replace OOI in equation (2) with call and put OOI and re-estimate the daily cross-sectional regressions. Panel A of Table 3 reports the results. The predictive powers of call OOI and put OOI do not completely subsume each other, supporting Ge, Lin, and Pearson (2016), who argue that synthetic long positions embed comparable or greater information content than short positions. In our sample, both call and put OOI maintain their stock return predictability across all 30-minute intraday intervals.

Next, Pan and Poteshman (2006) and Ge, Lin, and Pearson (2016) conclude that trades opening new positions are more informative than trades closing existing positions. To examine whether it extends to an intraday setting, we decompose total OOI into open OOI and close OOI:

$$OOI_Open_{i,d,t}=\frac{\left(VOL_{i,d,t}^{C,OB}+VOL_{i,d,t}^{P,OS}\right)-\left(VOL_{i,d,t}^{C,OS}+VOL_{i,d,t}^{P,OB}\right)}{\left(VOL_{i,d,t}^{C,OB}+VOL_{i,d,t}^{P,OS}\right)+\left(VOL_{i,d,t}^{C,OS}+VOL_{i,d,t}^{P,OB}\right)} \tag{4}$$

$$OOI_Close_{i,d,t}=\frac{\left(VOL_{i,d,t}^{C,CB}+VOL_{i,d,t}^{P,CS}\right)-\left(VOL_{i,d,t}^{C,CS}+VOL_{i,d,t}^{P,CB}\right)}{\left(VOL_{i,d,t}^{C,CB}+VOL_{i,d,t}^{P,CS}\right)+\left(VOL_{i,d,t}^{C,CS}+VOL_{i,d,t}^{P,CB}\right)} \tag{5}$$

Both open and close OOI are bullish measures. The distinction is that open (close) OOI uses only option trades that open new (close existing) positions. Panel B of Table 3 reports the results of the horse-race regression between open and close OOI. The return predictability of OOI is

primarily driven by open OOI. Consistent with Pan and Poteshman (2006) and Ge, Lin, and Pearson (2016), options trades that open new positions contain more information than trades that unwind existing positions.

To further support the conclusions above, we decompose total OOI into deeper categories:

$$OOI_Call_Open_{i,d,t}=\frac{VOL_{i,d,t}^{C,OB}-VOL_{i,d,t}^{C,OS}}{VOL_{i,d,t}^{C,OB}+VOL_{i,d,t}^{C,OS}},\ OOI_Call_Close_{i,d,t}=\frac{VOL_{i,d,t}^{C,CB}-VOL_{i,d,t}^{C,CS}}{VOL_{i,d,t}^{C,CB}+VOL_{i,d,t}^{C,CS}} \quad (6)$$

$$OOI_Put_Open_{i,d,t}=\frac{VOL_{i,d,t}^{P,OB}-VOL_{i,d,t}^{P,OS}}{VOL_{i,d,t}^{P,OB}+VOL_{i,d,t}^{P,OS}},\ OOI_Put_Close_{i,d,t}=\frac{VOL_{i,d,t}^{P,CB}-VOL_{i,d,t}^{P,CS}}{VOL_{i,d,t}^{P,CB}+VOL_{i,d,t}^{P,CS}} \quad (7)$$

where call (put) open and close OOI are bullish (bearish) measures, respectively. Panel C of Table 3 reports the results, which are consistent with the findings above: call open OOI exhibits the strongest predictive power across most intervals, followed by put open OOI. Trades that unwind existing positions contain limited information.

3.3. Options Leverage

Cao, Chen, and Griffin (2005), Pan and Poteshman (2006), and Ge, Lin, and Pearson (2016) find that the predictive power of signed option volume is stronger for high-leverage options such as OTM options, while Johnson and So (2012) and Hu (2014) demonstrate that the predictability is stronger for low-leverage options such as ATM and ITM options, owing to the higher transaction costs of OTM options. In this section, we conduct a heterogeneity analysis to examine the predictive power of OOI conditional on option moneyness.

We assign each option to one of three moneyness categories based on its strike price relative to the underlying stock prices, following the methodology provided by the ISE. A call (put) option is classified as OTM if the ratio of strike to stock price is between 1.05 and 1.20 (0.80 and 0.95), as ITM if this ratio is between 0.80 and 0.95 (0.80 and 0.95), and as ATM if this ratio is between 0.95 and 1.05. We then construct OOIs using OTM, ATM, and ITM option contracts only, and re-

estimate the Fama and MacBeth (1973) regression (2) for each moneyness category, respectively.

Panel A of Table 4 shows that OTM OOIs have the most significant predictive power and displays an intraday pattern akin to baseline results across intraday intervals, especially near the market close. These results are consistent with Cao, Chen, and Griffin (2005), Pan and Poteshman (2006), and Ge, Lin, and Pearson (2016), who emphasize the demand for leverage by informed options traders. This finding also supports Easley, O'Hara, and Srinivas (1998), who hypothesize that informed traders prefer OTM option contracts for their greater profit potential. In Panels B and C, ATM and ITM OOIs also exhibit predictive power in certain intraday intervals, e.g., near the market open and close, but generally have lower magnitudes than OTM OOI. These results are consistent with Johnson and So (2012) and Hu (2014), who argue that informed traders also weigh transaction costs against leverage benefits when selecting option contracts.

3.4. Investor Type

Market microstructure theory predicts that informed traders may split their orders over time to better camouflage their information among uninformed traders (e.g., Kyle, 1985; Easley and O'Hara, 1987). Figure 1 also documents that more than 65% of options trading volume originates from small customers. Taken together, we hypothesize that the predictive power of OOI is driven primarily by options trades from small customers.

Panel A of Table 5 reports the Fama and MacBeth (1973) regression results using OOI constructed by small-customer trades only. The statistical significance and economic magnitude are comparable to the baseline results in Table 2, suggesting that the predictive power of OOI stems primarily from small-size trading. Middle customer OOI (Panel B), large customer OOIs (Panel C), and firm OOIs (Panel D) exhibit limited predictive powers in some intervals only. The predictive power of small-size trades is consistent with the order splitting of informed traders (Kyle,

1985; Easley and O'Hara, 1987) or the surge of retail options investors (Eaton et al., 2026).

3.5. Stock Returns Predictability of Intraday OOI: Opening, Midday, and Closing

So far, we have shown that 30-minute OOI can positively predict next-day stock returns. To facilitate our investigation of the economic drivers of this predictability, we aggregate options trading into three broader time intervals. Following Hu et al. (2025), we partition the trading day into three intervals: near the open (9:30 a.m. to 11:30 a.m.), midday (11:30 a.m. to 2:00 p.m.), and near the close (2:00 p.m. to 4:00 p.m.). We re-calculate the corresponding OOIs and examine their predictive power for future stock returns using equation (2).

Panel A of Table 6 presents the Fama and MacBeth (1973) regression results. Consistent with Table 2, the coefficient of OOI is significantly positive across all three intervals, while the OOI near the close has the strongest predictive power. A one-standard-deviation increase in OOI near the open, midday, and near the close predicts, respectively, a 1.960 bps (*t*-stat. = 6.82; 5.06% per annum), 2.645 bps (*t*-stat. = 9.26; 6.89% per annum), and 3.605 bps (*t*-stat. = 12.61; 9.51% per annum) increase in next-day stock returns. This pattern persists when we regress next-day stock returns on three OOIs simultaneously, as shown in column (1) of Panel B, Table 6.

A concern with the Fama and MacBeth's (1973) two-stage regression is that the results may be driven by exorbitant coefficients on particular dates. As a robustness check, we plot the daily cross-sectional regression coefficients for three OOIs in Figure 3. The majority of (significant) coefficients lie above zero without abnormal spikes, ruling out the outlier effects and confirming that the positive return predictability of intraday OOI is persistent throughout our sample period.

The concentration of informed options trading near the market close appears to contradict the aggressive trading model by Holden and Subrahmanyam (1992) and the empirical evidence in the short-sale market by Hu et al. (2025). To examine whether this intraday pattern is stable over time,

we use the commencement of COVID as a cutoff and split the full sample into two subperiods: July 2017 to December 2019, and January 2020 to December 2022. We separately estimate the regression in column (1) of Panel B for two subperiods; columns (2) and (3) of Panel B, Table 6, report the results. Before COVID-19, the predictive power of OOI near the market open is stronger than the midday interval and comparable to that of the closing interval. After the onset of the pandemic, the OOI near the market close has the strongest predictive power, while the predictive power of OOI near the market open diminishes, transforming the intraday pattern from a U-shape to a J-shape. Figure OA4 illustrates this shift clearly. This structural change highlights a regime shift in option market structure in recent years, especially after the pandemic.

To test whether the predictive power of OOI persists or reverses over longer horizons, we examine the stock return predictability over the next 4 weeks. Panel C of Table 6 reports the results. The predictive power of OOI across all three intervals holds for future stock returns to at least four weeks ahead, while the magnitude and significance of the coefficients gradually diminish. The long-term predictive power of OOI is consistent with Kyle's (1985) model, in the sense that informed investors gradually trade on long-lived information, leading to long-run predictive power.

Out-of-sample analyses. Thus far, we have presented extensive evidence establishing the predictive power of OOI for future stock returns. To assess out-of-sample forecast performance, we designate July 2017 to December 2020 as the training subsample and January 2021 to December 2022 as the testing subsample and conduct two out-of-sample analyses.[16] Following Gu, Kelly, and Xiu (2020), we first calculate the out-of-sample R^2 as

$$R^2_{OOS} = 1 - \frac{\sum_{(i,t)\in\mathcal{T}} \left(Ret_{i,t+1} - \widehat{Ret}_{i,t+1}\right)^2}{\sum_{(i,t)\in\mathcal{T}} Ret^2_{i,t+1}} \tag{8}$$

[16] Our results remain similar with an alternative split of training and testing set, such as July 2017 to December 2021 as training subsample and January 2022 to December 2022 as testing subsample.

where $\mathcal{T}$ denotes the testing subsample. $Ret_{i,t+1}$ and $\widehat{Ret}_{i,t+1}$ are the realized and predicted returns of stock i at time t+1, respectively. A positive R^2_{OOS} indicates that the prediction model outperforms the benchmark out of sample.[17]

Next, to compare the forecast performance across intraday intervals, we construct the Diebold and Mariano (1995) test statistic, $DM_{12} = \bar{d}_{12}/\hat{\sigma}_{d_{12}}$, where $\bar{d}_{12}$ and $\hat{\sigma}_{d_{12}}$ are the time-series average and Newey-West (1987) robust standard error with 4 lags of d_{12}, respectively. The difference in predicted errors, d_{12}, is defined as

$$d_{12,t+1} = \frac{1}{n_{\mathcal{T},t+1}} \sum_{i=1}^{n_{\mathcal{T},t+1}} \left(\left(\hat{e}^{(1)}_{i,t+1}\right)^2 - \left(\hat{e}^{(2)}_{i,t+1}\right)^2 \right) \tag{9}$$

where $n_{\mathcal{T},t+1}$ denotes the number of stocks at time t+1 in the testing subsample, and $\hat{e}^{(1)}_{i,t+1}$ and $\hat{e}^{(2)}_{i,t+1}$ denote the prediction errors for stock i at time t+1 using intraday interval (1) and (2), respectively. A significantly positive DM_{12} indicates that model (2) outperforms model (1).

Panel D of Table 6 reports the out-of-sample R^2 and the Diebold-Mariano test statistics for the three intraday intervals. Consistent with the in-sample results, OOI near the stock market close continues to generate the highest out-of-sample R^2. Although there is no statistical difference in forecast performance between OOI near the market open and midday, OOI near the market close yields the best out-of-sample forecast performance.

4. Tests of Informed Trading

In this section, we conduct a series of tests examining the informed trading channel underlying the predictive power of intraday OOI. First, we study the relationship between informed options

[17] An alternative definition of out-of-sample R^2 is demeaning stock returns in the denominator with the historical mean of stock returns. As suggested by Gu, Kelly, and Xiu (2020), comparing against a naïve forecast of zero is more appropriate than against the historical mean when analyzing individual stock returns. Indeed, comparing against the historical mean still generates a positive but smaller out-of-sample R^2 in our sample.

traders and other well-documented types of informed traders: (i) institutional investors; (ii) short sellers; and (iii) insiders. Second, we explore two channels through which informed trading drives the predictive power of OOI: the liquidity timing channel and the information channel.

4.1. Informed Options Traders and Other Types of Informed Traders

The comovement between options trading and other forms of informed activity offers an alternative explanation for the predictive power of OOI. If the predictive signal in OOI simply reflects the trading behavior of institutional investors, short sellers, or insiders, controlling for these channels should weaken or eliminate the predictive power of OOI. This test allows us to assess whether OOI contains unique information or merely coincides with other informed trading.

First, we investigate three categories of 13F institutional investors. Bushee (1998) classifies institutional investors into three types based on their past investment patterns: transient investors, dedicated investors, and quasi-indexers. Transient investors have the highest trading turnover, the greatest use of momentum strategies, and relatively high diversification. Dedicated investors have a high concentration of ownership, low turnover, and near-zero momentum. Quasi-indexers have high diversification, low turnover, and low momentum. This classification is widely used in the institutional ownership literature (e.g., Wermers, 2000; Gibson, Safieddine, and Sonti, 2004).

We construct ownership for each institutional type, defined as the sum of shares held by all institutions of that type divided by total shares outstanding. To investigate whether the predictive power of OOI relates to institutional investors, we define three indicators, *TRA*, *DED*, and *QIX*, which respectively equal one if the transient, dedicated, and quasi-indexer ownership is above the quarterly cross-sectional median, and zero otherwise. We then interact OOI over three intraday intervals with each indicator to predict next-day stock returns:

$$\begin{aligned} Ret_{i,d+1} = {} & \alpha + \beta_1 \times TRA_{i,d} \times OOI_{i,d,t} + \beta_2 \times DED_{i,d} \times OOI_{i,d,t} + \beta_3 \times QIX_{i,d} \times OOI_{i,d,t} \\ & + \beta_4 \times OOI_{i,d,t} + \beta_5 \times TRA_{i,d,t} + \beta_6 \times DED_{i,d,t} + \beta_7 \times QIX_{i,d,t} + \psi' X_{i,d,t} + \varepsilon_{i,d+1} \end{aligned} \quad (10)$$

Panel A of Table 7 presents the Fama and MacBeth (1973) regression results of equation (10). Transient institutional investors significantly amplify the predictive power of OOI over three intraday intervals, with positive and significant coefficients of 1.109 (t-stat. = 2.38), 0.871 (t-stat. = 1.71), and 1.190 (t-stat. = 2.28), respectively, while the coefficient of OOI per se remains significant. This result suggests that part of OOI's predictive power derives from sophisticated institutions that actively participate in short-term and intraday informed trading.

In contrast, quasi-indexers significantly dampen the return predictability of OOI near the market close, as indicated by a negative interaction coefficient of −1.553 (t-stat. = −2.65). As we discuss in Section 4.2 later, the predictive power of OOI is lower when stock liquidity conditions is more favorable, which explains the weakening effect of quasi-indexers, who typically conduct uninformed stock trading near the market close to minimize their tracking errors. Finally, the interactions with dedicated ownership (*DED*) have limited predictive power for next-day stock returns. Dedicated investors hold large, long-term positions and are therefore unlikely to participate in intraday informed options trading or to predict short-term stock returns.

Recent studies document another important type of informed trader in the equity market: short sellers. Muravyev, Pearson, and Pollet (2025a) show that measures based on implied volatilities predict stock returns because they effectively capture the stock borrow fee. Muravyev, Pearson, and Pollet (2025b) find that asset pricing anomalies in the equity market are primarily driven by limits to arbitrage induced by short-sale constraints. Hu et al. (2025) examine the stock return predictability of intraday short-selling flow and find a stronger effect near the market open. Johnson and So (2012) suggest that the options market is an attractive venue for informed traders facing short-sale constraints. To examine the relation between options traders and short sellers, we collect stock-day level borrower concentration (BC) from Markit. A larger number of borrowers (higher

competition) decreases the value of BC from 1 to 0.[18] We define an indicator variable, *HighBC*, which equals one if a firm's BC is above the daily cross-sectional median, and zero otherwise. In other words, *HighBC* = 1 identifies hard-to-borrow stocks. To investigate the effect of short sellers on the predictive power of OOI, we interact OOI over three intraday intervals with *HighBC* to predict next-day stock returns.

Panel B of Table 7 presents the Fama and MacBeth (1973) regression results. When short-sale constraints intensify near the market close, the predictive power of OOI is significantly stronger, while the coefficient of OOI remains significant. This result suggests that short sellers migrate to the options market to bypass short-sale constraints in the equity market when short-sale cost is higher (i.e., the stock is hard-to-borrow), consistent with Johnson and So (2012). This result also reconciles the weaker predictive power of short-selling flows near the market close documented in Hu et al. (2025). As informed short sellers typically surround the market open, the short-sale market near the close becomes less competitive, incentivizing constrained informed traders to trade options instead, either long puts or short calls. Consistent with conventional wisdom, the significantly positive coefficients of *HighBC* indicate that stocks targeted by more short sellers would experience lower next-day returns.

Finally, we focus on insiders (Cohen, Malloy, and Pomorski, 2012; Bondarenko and Muravyev, 2025). We are particularly interested in the intraday options trading activities in the presence of insider information. We define $Insider_{i,d}$ as an indicator that equals one if insiders of stock *i* conduct options transactions on day *d*, and zero otherwise.[19] Panel C of Table 7 presents the results after intersecting *Insider* with OOI. We find no effect of insider trading on options

[18] The definitions of borrower concentration and indicative borrow fee are provided by Markit.
[19] We collect information on open market derivative transactions from Table 2 Form 4 filings, which document changes in insiders' derivative positions. We retain call and put options, warrants, employee stock options, and director stock options. We discard option expirations and swap transactions, and records with cleansing codes S or A.

informed trading: on insider transaction days, the predictive power of OOI is statistically indistinguishable from that on normal days. This finding suggests that insider options trading contains no private information on future stock returns. It is also consistent with the view that insider transactions typically do not involve options because of the heightened risk of detection (Collin-Dufresne, Fos, and Muravyev, 2021). Importantly, the coefficients of OOI remain significant after controlling for insider trading.

Taken together, we find that informed options traders substantially overlap with, yet provide distinct information beyond other types of informed traders, such as sophisticated institutional investors and short sellers. Moreover, the options market becomes more attractive to informed traders when they face greater constraints in the equity market, especially near the market close, such as less regular and uninformed passive trading and higher short-selling costs.

4.2. Liquidity Timing

In this section, we investigate whether informed options trading varies with options liquidity conditions as proposed by Collin-Dufresne and Fos (2016) and Kacperczyk and Pagnotta (2019), as well as with underlying stock liquidity as modeled by Easley, O'Hara, and Srinivas (1998).

The most common context for elevated options trading volume is option expiration days, the third Friday of each month. A large fraction of expiration-week option volume is attributable to rolling over expiring positions into later expiring contracts. Although this rollover-induced trading contains no information,[20] it provides a noisy trading environment for informed options traders to conceal their information. Among these Fridays, derivative markets experience the heaviest expiry activity on "Triple Witching" days, which fall on the third Friday of March, June, September, and December, when different types of contracts expire simultaneously: stock market index futures,

[20] In this paper, we only select option contracts with day-to-expiration (DTE) to be greater than or equal to 10 days. We find no predictive power of OOI constructed by option contracts with DTE less than 10 days.

stock market index options, and stock options. Expiration days provide an ideal vehicle to examine whether informed options traders time the liquidity in the options market. We define two indicators, $3rdFriday_d$ and $TripleWitching_d$ that equal to one if day d is the third Friday of each month and a "Triple Witching" day, respectively. The following regression augments equation (2) by interacting OOI with two indicators:

$$Ret_{i,d+1} = \alpha + \beta_1 \times TripleWitching_d \times OOI_{i,d,t} + \beta_2 \times 3rdFriday_d \times OOI_{i,d,t} + \beta_3 \times OOI_{i,d,t} + \beta_4 \times TripleWitching_d + \beta_5 \times 3rdFriday_d + \psi' X_{i,d,t} + \varepsilon_{i,d+1} \quad (11)$$

If informed options traders time the liquidity conditions in option market, we expect both β_1 and β_2 to be positive, especially near the market close. Panel A of Table 8 presents the results. Consistent with our expectation, the liquidity-timing behavior is most pronounced near the market close. The sensitivity of OOI to future stock returns increases by 2.68% (=0.092/3.428) on the third Friday, and 5.16% (=(0.092+0.085)/3.428) on the "Triple Witching" days.

It is not surprising that liquidity timing on expiration days appears only near the market close, as single-name equity option contracts are p.m. settled, i.e., the final value of an option contract is determined by the underlying stock's closing price on the expiration day. Next, we explore liquidity timing across options and stock markets under more general circumstances. We use the natural logarithm of 30-minute options and stock trading volume to proxy for intraday liquidity conditions in the options and stock markets, respectively. Panel B of Table 8 presents the results. We find that informed options traders appear to time liquidity in the options market throughout the trading day. The predictive power of OOI is stronger when options market liquidity is higher at the corresponding intraday intervals.

More interestingly, informed options traders also jointly time the liquidity conditions in underlying equity market, but in the opposite direction: when underlying stock liquidity is better, the predictive power of OOI becomes weaker. Two possible explanations account for this pattern.

First, informed traders may prefer to trade directly in the stock market when equity market conditions are favorable, as suggested by Easley, O'Hara, and Srinivas (1998). Second, some options trading strategies, such as covered call/put or delta-hedging, may create offsetting price pressure on underlying stocks, which mechanically drive this negative effect.

Trading volume, however, may merely reflect disagreement rather than liquidity. Thus, we use percentage bid-ask spreads to proxy for liquidity conditions as a robustness check. Due to data limitations, we use the end-of-day national best bid-ask spread. This figure is then scaled by the closing price into a unitless (percentage-like) measure. A lower bid-ask spread indicates better liquidity conditions. We average option bid-ask spreads across eligible contracts to obtain stock-day level variable.[21] Consistent with Panel B, Panel C of Table 8 shows that the predictive power of OOI is significantly higher when options liquidity is better relative to underlying stocks.

Taken together, these findings indicate that informed options traders actively time the liquidity across both option and underlying stock markets. The predictive power of OOI strengthens when the liquidity in option market is better relative to underlying stock market, particularly near the stock market closing bell. We provide comprehensive empirical tests of the theoretical implications of Easley, O'Hara, and Srinivas (1998) using more recent sample period.

4.3. Private and Processed Public Information Channel

Prior studies document stronger predictive power of options trading around firm-specific news releases. For example, Lin and Lu (2015) find that the predictive power of option-implied volatility increases around analyst events. Bergsma et al. (2020) document stronger predictive power of the O/S ratio in the first half hour when an earnings announcement is released last overnight. Chordia,

[21] Following previous studies (e.g., Cao et al., 2026), we only retain option contracts with positive bid prices, quote midpoints greater than \$0.125, bid-ask spreads lower than the minimum tick size (i.e., \$0.10 for option prices above \$3, and \$0.05 otherwise). We exclude option contracts violating no-arbitrage conditions, i.e., $S \geq C \geq \max(0, S - Ke^{-rt})$ for call options with price C, and $S \geq P \geq \max(0, K - S)$ for put options with price P, where S denotes underlying stock closing prices, K denotes strike prices, r denotes risk-free rate, and t denotes time to expiration.

Lin, and Xiang (2021) propose that corporate news increases the predictive power of risk-neutral skewness for subsequent stock returns. In this section, we first examine the sensitivity of OOI's predictive power to firm-specific news through the *private* information channel.

Among all firm-level news, the most important information events are earnings and analyst-related news (Hu et al., 2025). We obtain data on all firm-specific news from the Full Edition RavenPack 1.0 database. Earnings and analyst-related news are labelled by the GROUP field as "earnings," "revenues," and "analyst-ratings." We require the RELEVANCE score to be 100 and the EVENT SIMILARITY DAYS to be 365 days, to ensure the news is not stale or outdated.[22] Because OOI is a bullish measure and its predictive power mainly stems from call options, we select positive news with EVENT SENTIMENT SCORE greater than zero, where EVENT SENTIMENT SCORE is a stock-event sentiment score between −1 and +1, computed by RavenPack using its proprietary algorithm. As Weinbaum et al. (2023) find that options signed volume responds only to non-scheduled news, we further restrict the sample to non-scheduled news with the taxonomy label SCHEDULED equals zero.[23] Specifically, we focus on options trading that occurs before positive non-scheduled corporate news will be released between day d's market close and day d+1's market open and refer to this as the private information channel. We add an interaction term between the OOI and indicator variable capturing firm-specific news:

$$Ret_{i,d+1} = \alpha + \beta_1 \times POS_NSCH_{i,d} \times OOI_{i,d,t} + \beta_2 \times OOI_{i,d,t} + \beta_3 \times POS_NSCH_{i,d} + \psi' X_{i,d,t} + \varepsilon_{i,d+1} \quad (12)$$

where $POS_NSCH_{i,d}$ is an indicator that equals one if at least one positive non-scheduled earnings or analyst-related news will be released after day d's market close and before d+1's market open. We also include the interaction of SOI with *POS_NSCH* to control informed trading in the

[22] Hafez (2009) finds that 80% of all news stories simply add noise.

[23] Our results remain robust when further interacting OOI with other types of news, i.e., negative non-scheduled, positive scheduled and negative scheduled news.

underlying stock market. The predictability of OOI is captured by $\beta_1 \times POS_NSCH_{i,d} + \beta_2$, where β_1 estimates the predictability of OOI conditional on firm-specific news releases. If the information flow from options to the stock market contains such private information, we expect $\beta_1 > 0$.

Table 9 presents the results. In Panel A, we examine how intraday OOI behaves when earnings or analyst-related news will be released after today's market close. We regress OOI of three intraday intervals on the news indicator $POS_NSCH_{i,d}$. We find that only OOI near the market close leads to a potential news release. In Panel B, we regress $POS_NSCH_{i,d}$ on OOI from each of the three intervals to examine whether OOI predicts a subsequent news release, and find consistent results. Panel C presents the Fama and MacBeth (1973) regression results of equation (12). Similarly, we find a significant β_1 only near the market close, while options trading near the market open and midday does not contain private information about earnings or analyst ratings. Taken together, our results suggest that informed options traders prioritize minimizing price impact over preserving their information advantage. Just before the release of earnings or analyst-related news, they exploit better liquidity conditions near the market close to achieve a smaller price impact, despite the decay of their private information advantage.

Jin, Livnat, and Zhang (2012) argue that options traders have a better ability to process public information than equity traders. We test this hypothesis by examining options trading after corporate news released between day $d-1$'s market close and day d's market open and refer to this as the *processed public* information channel. We replace $POS_NSCH_{i,d}$ in equation (12) with $POS_NSCH_{i,d-1}$, an indicator that equals one if at least one positive non-scheduled news was released after day $d-1$'s market close and before day d's market open. Panel D of Table 9 presents the Fama and MacBeth (1973) regression results. When positive non-scheduled earnings or analyst-related news was released last overnight, informed options traders respond immediately

after the market open. This pattern dissipates as the trading day approaches the market close. Our results contribute to prior literature by highlighting the heterogeneous behavior of informed options traders within a trading day: they process and impound public information immediately after the market open, then turn to trading on private information near the market close to benefit from reduced price impact.

4.4. Macroeconomic Events

Beyond firm-specific news, we next turn to important macroeconomic news, including FOMC meetings, VIX, and interest rate shocks, for two reasons. First, macro research pursues the joint dynamics of macro news and asset prices (Duffee, 2023), and macro news is often associated with changes in interest rates, such as Treasury yields, which intuitively affect asset valuations. Second, Liu, Peng, and Tang (2023) find that macro news may crowd out retail investors' attention to earnings announcements and reduce earnings announcement returns. This shift in focus calls for attention to macroeconomic news in the study of informed options trading.

We begin with FOMC meetings. The Federal Reserve's monetary policy committee holds scheduled meetings to make key decisions on interest rates and other policy tools. Lucca and Moench (2015) show that a substantial portion of the equity risk premium is earned in the short window preceding FOMC announcements, a phenomenon known as the pre-FOMC drift. Recently, Acosta et al. (2025) find that press conferences have become the primary source of monetary policy news, often overshadowing the policy statement itself. To study how the predictive power of OOI varies around these events, we introduce three indicators and interact them with OOI. These indicators capture different phases of monetary policy communication and allow us to examine whether informed options trading responds to macroeconomic events:

$$\begin{aligned} Ret_{i,d+1} = {} & \alpha + \beta_1 \times FOMC_d^{-1} \times OOI_{i,d,t} + \beta_2 \times FOMC_d^{0} \times OOI_{i,d,t} + \beta_3 \times Press_d \times OOI_{i,d,t} \\ & + \beta_4 \times OOI_{i,d,t} + \beta_5 \times FOMC_d^{-1} + \beta_6 \times FOMC_d^{0} + \beta_7 \times Press_d + \psi' X_{i,d,t} + \varepsilon_{i,d+1} \end{aligned} \quad (13)$$

where $FOMC_d^{-1}$, $FOMC_d^0$, and $Press_d$ are indicators that equal one if day d is, respectively, the day before the FOMC meeting, the (first) day of the FOMC meeting, and the press conference day, and zero otherwise.[24] Panel A of Table 10 reports the Fama and MacBeth (1973) regression results of equation (13). On the day prior to the FOMC meeting, β_1 is significantly positive for opening and closing intervals, suggesting that options trading in these intervals already incorporates macro information before FOMC meetings. On the FOMC meeting day, informed options trading is further intensified in all three intervals. After the FOMC announcements at 2:00 p.m., the significantly positive β_2 for the closing period indicates that informed options traders have processed and impounded the public macro information. These information channels are further supported by the significantly positive β_3 for opening and midday periods in press conference days.

FOMC meeting is a special event typically accompanied by sharp interest rates and uncertainty shocks. To broaden the results, we construct a daily interest rate shock ($|IR_d|$) as the absolute change in the 1-year nominal zero-coupon bond (ZCB) yield following Duffee (2023).[25] Analogously, we construct a daily uncertainty shock ($|VIX_d|$) as the absolute change in VIX.[26] We interact these variables with OOI to investigate options trading when uncertainty and interest rate risks elevate. In Panel B of Table 10, the coefficients of interactions in all three intervals are significantly positive, indicating that an elevated uncertainty shock increases the predictive power of OOI. In column (3), a 1 index-point change in VIX induces a 5.67% (=0.185/3.260) increase in the predictive power of OOI. In Panel C of Table 10, we replace $|VIX_d|$ with $|IR_d|$ and find significantly positive coefficients of interactions over midday and near the close. In column (3), a 1% increase in $|IR_d|$ induces a 6.94% (=0.223/3.215) increase in the predictive power of OOI.

[24] FOMC meeting schedules can be found on Federal Reserve website.
[25] The downloadable sheet for yield can be found on the Federal Reserve website. We also use the 10-year and 30-year nominal ZCB yield for robustness checks and obtain similar results.
[26] CBOE VIX historical data is made available on the CBOE official website.

5. Additional Tests

5.1. Stock Returns Predictability of OBPC

Beyond OOI, several papers document the significant predictive power of other option volume-based measures for future stock returns. For example, Pan and Poteshman (2006) show that the put-call volume ratio negatively predicts cross-sectional stock returns. More recently, Weinbaum et al. (2023) document that options trading activity conveys information about firm news using the open-buy put-call ratio. In this section, we extend our baseline analysis by replacing OOI with the open-buy put-call ratio, defined as $OBPC_{i,d,t} = VOL^{P,OB}_{i,d,t}/(VOL^{P,OB}_{i,d,t}+VOL^{C,OB}_{i,d,t})$.

A higher OBPC indicates a relatively higher volume of longing put options than call options. Thus, we expect OBPC to negatively predict stock returns. Panel A of Table OA5 examines the predictive power of OBPC for next-day stock returns over three intraday intervals. In all three columns, the coefficient of OBPC is significantly negative, consistent with prior literature. The predictive power of OBPC is also the strongest near the market close. In column (3), a one-standard-deviation increase in OBPC predicts a 2.667 bps (t-stat. = −7.03, 6.95% per annum) decrease in next-day stock returns.

5.2. Jump Predictability of OOI

The conventional wisdom typically attributes stock price jumps to news releases. Aït-Sahalia, Li, and Li (2025), however, document a puzzle that most stock price jumps are not accompanied by public news releases. In this section, we demonstrate that options trading induces stock price pressure, which triggers stock price jumps the next trading day. Following Aït-Sahalia, Li, and Li (2025), we detect price jumps in each 30-minute interval using the following formula: $\left|Ret_{i,d+1,t}\right| > c\times RV_{i,d+1,t}\times\sqrt{\Delta}$, where $Ret_{i,d+1,t}$ is the 30-minute returns of stock i on day d+1 over

time interval $(t-1, t]$. $RV_{i,d+1,t}$ is the realized volatility of 1-minute returns of stock i on day $d+1$ over time interval $(t-1, t]$. Δ denotes the length of the time interval between $t-1$ and t. c denotes the price jump size, defined as the multiple of the standard deviation of stock returns. We aggregate c over thirteen 30-minute intervals to obtain the daily stock jump size on day $d+1$, $JumpSize_{i,d+1}$. We replace stock returns on the left-hand side of equation (2) by jump size, and Panel B of Table OA5 presents the results. We find that OOI predicts larger price jumps next trading day. Consistent with the baseline pattern, the predictive power of OOI near the market close is the strongest.

5.3. Crash Predictability of OOI

With the global COVID-19 pandemic and other "unexpected" events since the 2008 financial crisis, sudden plunges in various financial markets have become more frequent.[27] It is therefore valuable to identify predictors of crash risks, especially at the intraday level. Following Amaya et al. (2015), we define annualized realized skewness as a proxy for crash risks: $RSkew_{i,d+1} = \sqrt{252}\times\sqrt{13}\times \sum_{j=1}^{13} Ret_{i,d+1,j}^3 / \left(\sum_{j=1}^{13} Ret_{i,d+1,j}^2\right)^{3/2}$, where $Ret_{i,d+1,j}$ denotes the j-th 30-min return of stock i on day $d+1$. A higher realized skewness indicates a longer right tail and lower crash risks. We examine whether OOI predicts crash risks by replacing the dependent variable in equation (2) with next-day realized skewness. Panel C of Table OA5 presents the Fama and MacBeth (1973) regression results. We obtain significantly positive coefficients of OOI near the market close. In column (3), a one-standard-deviation increase in OOI predicts a 0.052 (t-stat. = 2.38) increase in realized skewness, i.e., lower crash risks.

[27] For example, the 7% decrease in the price of the S&P 500 index triggers the circuit breakers on March 9, 2020. The price of West Texas Intermediate (WTI) crude oil near-term futures once falls to a negative value on April 20, 2020. Due to the Russia-Ukraine conflicts and sanctions on Russia, the Russian Ruble plunge by 30% to the lowest recorded number on February 28, 2022. On March 9, 2022, Dow Jones, NASDAQ, and S&P 500 all decrease by more than 2%, among which S&P 500 index plummet by 2.95%, creating the lowest recorded one-day fall since 2020.

5.4. Robustness Checks

To further support the baseline results, we conduct a series of robustness checks. First, Muravyev, Pearson, and Pollet (2025a) show that option-priced measures predict stock returns because they effectively capture stock borrow fees, which are already known to predict stock returns. Is the predictive power of OOI also driven by the implied stock borrow fee? To address this question, we further control the indicative stock borrow fee in equation (2). Panel A of Table OA6 reports the results. After controlling the stock borrow fee, the forecasting power of every 30-minute OOI remains significant and presents a similar intraday pattern with the baseline results.

Although Fama and MacBeth (1973) regressions establish the cross-sectional predictability, some asset pricing literature (e.g., Lin and Lu, 2015) uses the pooled regression with fixed effects to control for unobservables. We check the robustness under this setup, and Panel B of Table OA6 presents the results. We include both stock and day fixed effects to control for time-invariant and time-variant factors. The predictive power of OOI remains robust across all intervals. To further support the cross-sectional predictive power of OOI, we use the industry fixed effects in lieu of stock fixed effects (Roberts and Whited, 2013) and obtain similar results in Panel C of Table OA6.

Finally, we repeat the baseline regressions using only CBOE and ISE data, respectively, and present the corresponding results in Panels D and E of Table OA6. First, our baseline results are primarily driven by informed options trading in CBOE, as we obtain a similar pattern of return predictability for *OOI_CBOE* compared to the baseline results. Second, for options trading in ISE, *OOI_ISE* presents no predictive power at most intraday intervals. However, the predictive power of *OOI_ISE* remains significant and sometimes is even stronger near the market close. Since both OOIs constructed by trades from CBOE and ISE have significant predictive power, combining data from these two exchanges better represents the entire options market.

6. Conclusions

We combine high-frequency signed options volume data from two major U.S. options exchanges to examine the intraday information flow between the options and equity markets. We find that all 30-minute options order imbalances positively predict next-day stock returns, with the strongest effects near the market close. This predictive power is particularly driven by call options trades opening new positions, OTM options, and small trades, consistent with theories of options leverage and order splitting. Informed options traders respond to public news near the market open and act on private information near the close. They also jointly time the liquidity in both option and underlying stock markets, favoring options when options liquidity is better relative to underlying stocks. OOI also forecasts stock price jumps and crash risk, highlighting the critical role of option market in incorporating information about downside tail risks.

Our study raises an important caveat regarding the evolving structure of today's options market. In recent years, options trading volume has increased substantially. We document a shift in the composition and dynamics of intraday options trading, particularly near the market close. Our findings highlight a notable pattern in informed trading activity in recent periods, suggesting that future research should pay close attention to options trading at the end of the trading day when studying information flow within the options market or to the stock market.

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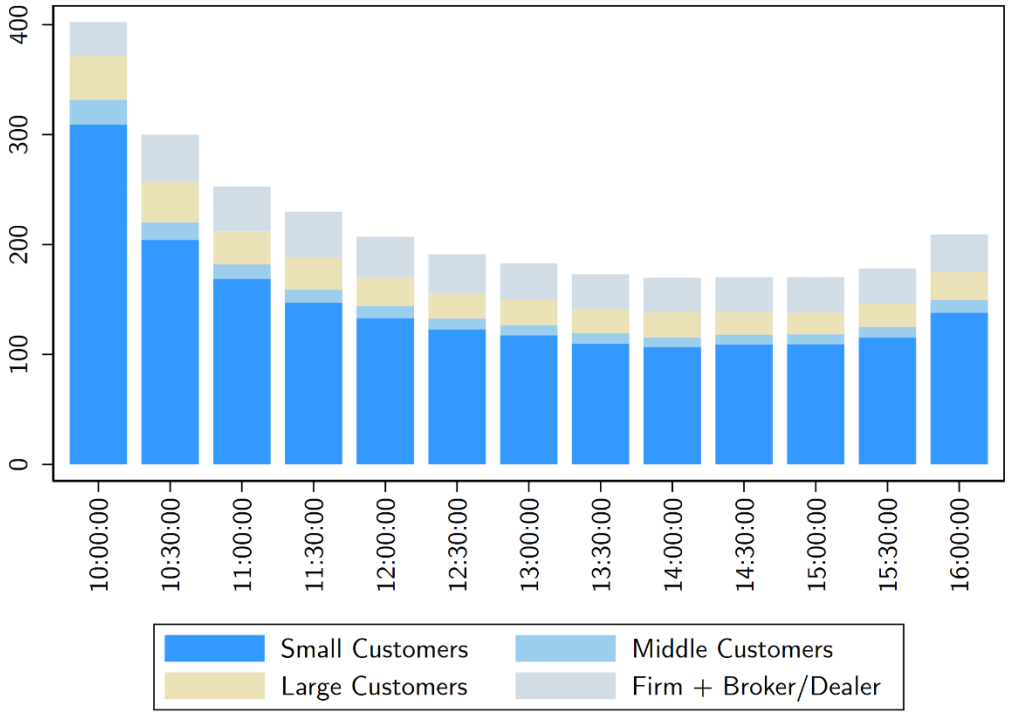


Figure 1. Intraday average of options trading volume.

This figure shows the average options trading volume of different investor types in the CBOE and ISE data across every 30-minute interval. The sample period spans from July 2017 to December 2022.

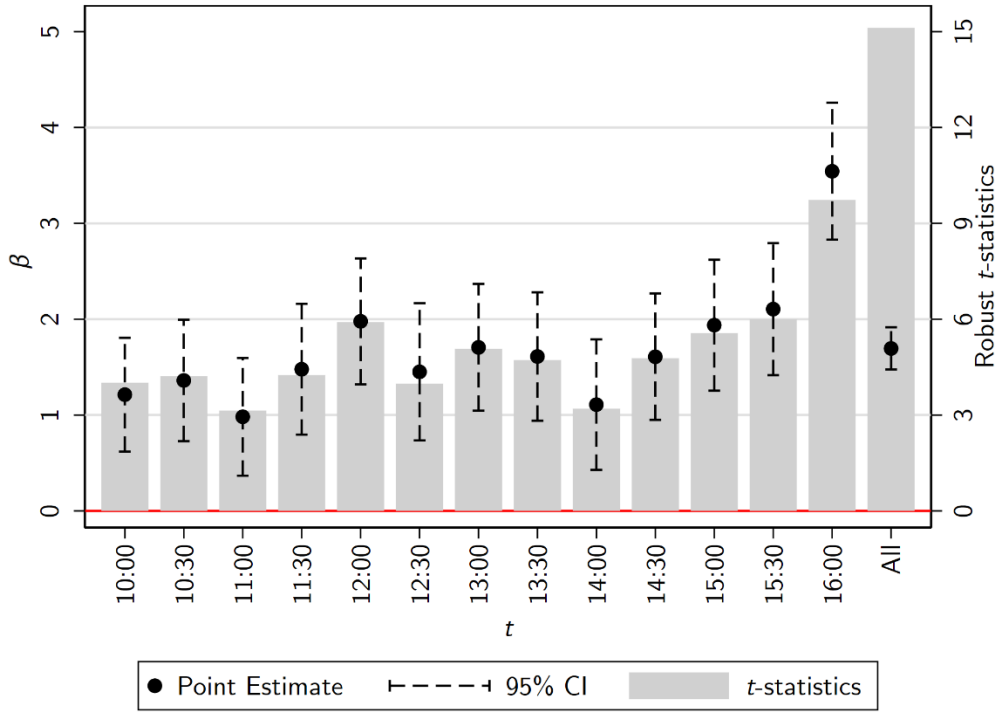


Figure 2. Intraday OOI coefficient, robust *t*-statistics, and 95% confidence interval.

This figure shows the $\widehat{\beta}$ with 95% confidence interval, and corresponding robust *t*-statistics shown in Table 2. The sample period is from July 2017 to December 2022.

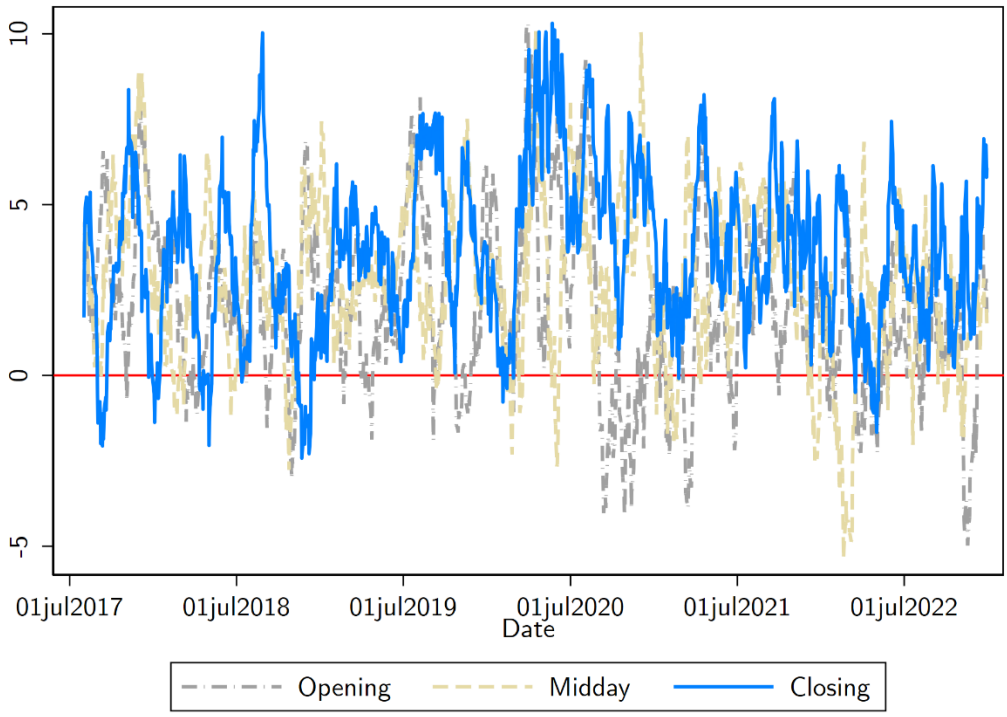


Figure 3. Daily coefficients of OOI in Fama-MacBeth (1973) regression.

This figure depicts the daily coefficients of OOI in Fama-MacBeth (1973) regression (with standardization). Opening, midday, and closing intervals consist of trading over [9:30,11:30), [11:30,14:00), and [14:00,16:00), respectively. We calculate the 20-day moving average of the coefficients to smooth the time series. The sample period is from July 2017 to December 2022.

Table 1. Descriptive Statistics

Panel A: Summary Statistics

	N	Mean	*SD*	Min	Q1	Median	Q3	Max
$Ret_{i,d+1}$	1,901,855	2.176	326.211	−1012.312	−152.866	0.000	150.602	1134.117
$OOI_Opening_{i,d}$	1,504,526	−0.020	0.643	−1.000	−0.500	0.000	0.409	1.000
$OOI_Midday_{i,d}$	1,388,680	−0.025	0.651	−1.000	−0.500	0.000	0.415	1.000
$OOI_Closing_{i,d}$	1,340,350	−0.008	0.655	−1.000	−0.500	0.000	0.455	1.000
$SOI_Opening_{i,d}$	1,727,148	−0.020	0.183	−0.551	−0.129	−0.018	0.091	0.497
$SOI_Midday_{i,d}$	1,725,495	−0.011	0.160	−0.485	−0.103	−0.010	0.082	0.439
$SOI_Closing_{i,d}$	1,726,815	−0.005	0.141	−0.415	−0.087	−0.004	0.078	0.391
$StockVol_Opening_{i,d}$	1,504,526	3.371	2.058	0.000	1.792	3.258	4.787	8.477
$StockVol_Midday_{i,d}$	1,388,680	3.299	2.014	0.000	1.792	3.178	4.682	8.360
$StockVol_Closing_{i,d}$	1,340,350	3.254	1.991	0.000	1.792	3.135	4.635	8.212
$OptionVol_Opening_{i,}$	1,729,198	12.326	1.369	9.048	11.399	12.294	13.213	15.844
$OptionVol_Midday_{i,d}$	1,729,091	12.104	1.287	8.957	11.242	12.081	12.939	15.392
$OptionVol_Closing_{i,d}$	1,728,531	12.535	1.218	9.454	11.725	12.532	13.342	15.535
$Spread_Opening_{i,d}$	1,729,211	0.137	0.212	0.007	0.021	0.059	0.154	1.285
$Spread_Midday_{i,d}$	1,729,182	0.077	0.122	0.004	0.013	0.032	0.083	0.774
$Spread_Closing_{i,d}$	1,729,129	0.065	0.101	0.005	0.012	0.028	0.068	0.649
$Turnover_{i,d}$	1,906,609	0.014	0.016	0.002	0.006	0.009	0.016	0.114
$Volatility_{i,d}$	1,905,499	0.502	0.334	0.107	0.275	0.412	0.623	1.976
$Size_{i,d}$	1,752,440	7.856	1.761	3.819	6.623	7.878	9.099	11.837
$BTM_{i,d}$	1,624,189	0.444	0.430	0.013	0.156	0.314	0.584	2.483
$Ret_{i,d-1}$	1,898,962	5.864	333.032	−1024.752	−151.891	2.375	154.739	1187.261
$Ret_{i,d-20,d-2}$	1,907,794	0.009	0.150	−0.398	−0.070	0.005	0.078	0.560
$Ret_{i,d-120,d-21}$	1,904,256	0.054	0.388	−0.681	−0.172	0.010	0.201	1.759

Panel B: Correlations

	OOI			*SOI*			AR(1)
	Opening	*Midday*	*Closing*	*Opening*	*Midday*	*Closing*	
$OOI_Opening_{i,d}$	1						0.0660
$OOI_Midday_{i,d}$	0.1158	1					0.0612
$OOI_Closing_{i,d}$	0.0912	0.1206	1				0.0623
$SOI_Opening_{i,d}$	−0.0306	−0.0261	−0.0251	1			0.0909
$SOI_Midday_{i,d}$	−0.0059	−0.0112	−0.0146	0.1897	1		0.1053
$SOI_Closing_{i,d}$	−0.0051	−0.0057	−0.0135	0.1310	0.2582	1	0.1212

Note. This table shows the descriptive statistics of unstandardized values for the main variables used in the baseline empirical specification. The sample period spans from July 2017 to December 2022. All unbounded and continuous variables are winsorized at the top and bottom 1% percentile. All variables are defined in Online Appendix A. Table OA1 lists the summary statistics of all other variables.

Table 2. The Stock Return Predictability of Intraday OOI

	(1) t = 10:00	(2) t = 10:30	(3) t = 11:00	(4) t = 11:30	(5) t = 12:00	(6) t = 12:30	(7) t = 13:00	(8) t = 13:30	(9) t = 14:00	(10) t = 14:30	(11) t = 15:00	(12) t = 15:30	(13) t = 16:00	(14) All
OOI	**1.213*** **	**1.361*** **	**0.982*** **	**1.478*** **	**1.978*** **	**1.451*** **	**1.706*** **	**1.611*** **	**1.109*** **	**1.608*** **	**1.938*** **	**2.105*** **	**3.543*** **	**1.695*** **
	(4.01)	**(4.22)**	**(3.14)**	**(4.25)**	**(5.91)**	**(3.98)**	**(5.07)**	**(4.72)**	**(3.20)**	**(4.78)**	**(5.56)**	**(6.00)**	**(9.73)**	**(15.12)**
SOI	−0.540	−0.699*	−0.406	−0.417	−0.527	−0.210	0.156	−0.372	0.461	−0.019	−0.068	−0.411	1.057***	−0.123
	(−1.57)	(−1.73)	(−1.19)	(−1.16)	(−1.33)	(−0.52)	(0.38)	(−0.87)	(1.14)	(−0.04)	(−0.17)	(−1.00)	(2.65)	(−0.89)
StockVol	−1.541***	−0.794*	−0.923**	−1.266***	−1.254***	−0.677	−0.880*	−1.415***	−0.942**	−0.281	−1.401***	−1.058**	−0.990**	−1.023***
	(−3.39)	(−1.91)	(−2.26)	(−2.98)	(−2.88)	(−1.49)	(−1.83)	(−3.10)	(−2.16)	(−0.59)	(−3.09)	(−2.12)	(−2.14)	(−4.81)
OptionVol	−2.848***	−3.321***	−3.838***	−2.926***	−3.495***	−3.811***	−3.752***	−3.765***	−3.533***	−4.031***	−3.680***	−4.139***	−4.757***	−3.798***
	(−3.19)	(−3.64)	(−4.39)	(−3.39)	(−3.99)	(−4.27)	(−4.11)	(−4.06)	(−3.90)	(−4.05)	(−4.07)	(−4.62)	(−5.62)	(−6.86)
Spread	−1.380**	−1.148**	−1.782***	−1.844***	−1.794***	−2.060***	−2.028***	−2.244***	−1.597***	−1.905***	−1.614***	−1.816***	−2.292***	−1.958***
	(−2.57)	(−2.12)	(−3.39)	(−3.40)	(−3.34)	(−3.75)	(−3.62)	(−4.25)	(−3.04)	(−3.55)	(−3.05)	(−3.32)	(−4.91)	(−5.67)
Turnover	0.627	0.613	0.633	−0.043	0.567	0.884	0.789	0.403	−0.192	0.301	0.335	0.470	0.514	0.467
	(0.67)	(0.61)	(0.58)	(−0.04)	(0.53)	(0.81)	(0.70)	(0.36)	(−0.17)	(0.25)	(0.30)	(0.43)	(0.50)	(0.72)
Volatility	−2.452**	−2.867**	−2.171*	−2.207*	−2.426*	−3.038**	−2.127	−1.706	−2.051	−2.937**	−3.057**	−3.009**	−2.789**	−2.552***
	(−2.04)	(−2.30)	(−1.72)	(−1.74)	(−1.89)	(−2.35)	(−1.54)	(−1.28)	(−1.56)	(−2.26)	(−2.27)	(−2.31)	(−2.14)	(−3.11)
Size	5.855***	5.749***	6.243***	5.517***	6.417***	5.831***	6.572***	6.657***	6.077***	6.205***	6.108***	7.117***	7.276***	6.234***
	(4.44)	(4.16)	(4.66)	(4.21)	(4.77)	(4.26)	(4.83)	(4.93)	(4.41)	(4.34)	(4.36)	(5.13)	(5.55)	(7.41)
BTM	1.847*	1.666	1.930*	1.439	1.592	1.433	1.943*	1.365	1.674	1.347	1.327	1.660	1.946*	1.641***
	(1.76)	(1.62)	(1.89)	(1.33)	(1.51)	(1.32)	(1.81)	(1.26)	(1.63)	(1.25)	(1.24)	(1.60)	(1.87)	(2.59)
Ret_{d-1}	−1.163	−0.887	−1.073	−1.140	−0.972	−1.527	−1.342	−2.186**	−1.819*	−0.963	−1.210	−1.724*	−0.955	−1.304**
	(−1.35)	(−0.97)	(−1.16)	(−1.19)	(−1.05)	(−1.56)	(−1.46)	(−2.31)	(−1.93)	(−0.96)	(−1.24)	(−1.86)	(−1.04)	(−2.25)
$Ret_{d-20,d-2}$	−0.385	−0.999	−0.774	−0.742	−1.334	−0.524	−0.589	−0.905	−0.552	−1.105	−0.629	−0.603	−0.702	−0.762
	(−0.37)	(−0.89)	(−0.69)	(−0.64)	(−1.17)	(−0.44)	(−0.51)	(−0.74)	(−0.46)	(−0.95)	(−0.53)	(−0.51)	(−0.63)	(−0.98)
$Ret_{d-120,d-21}$	2.607**	2.810**	2.391*	2.578*	2.630*	1.579	2.061	2.061	2.874**	2.380*	2.164	2.684**	2.172	2.365***
	(2.00)	(2.05)	(1.76)	(1.90)	(1.93)	(1.13)	(1.51)	(1.46)	(2.06)	(1.73)	(1.58)	(1.98)	(1.62)	(2.58)
Const.	−3.979	−4.010	−4.095	−4.007	−3.851	−4.245	−4.108	−3.403	−3.883	−4.391	−4.097	−4.414	−4.609	−4.267*
	(−1.05)	(−1.05)	(−1.09)	(−1.07)	(−1.03)	(−1.12)	(−1.09)	(−0.92)	(−1.04)	(−1.17)	(−1.11)	(−1.19)	(−1.23)	(−1.77)
# of Obs.	970,163	820,713	786,048	750,582	723,595	697,570	677,857	665,384	664,244	669,809	678,413	711,093	806,912	9,622,345
Adj. R^2	0.1104	0.1169	0.1175	0.1195	0.1192	0.1203	0.1206	0.1223	0.1201	0.1197	0.1190	0.1168	0.1130	0.1180

Note. This table shows the following Fama and MacBeth (1973) regression results of OOI predicting future returns: $Ret_{i,d+1}=\alpha+\beta\times OOI_{i,d,t}+\psi' X_{i,d,t}+\varepsilon_{i,d+1}$, where $Ret_{i,d+1}$ denotes the close-to-close returns of stock i on day d+1 in basis points. $OOI_{i,d,t}$ is 30-minute options order imbalance using the trading volume of options underlying stock i on day d from time t−1 to t. $X_{i,d,t}$ denotes a battery of control variables: $SOI_{i,d,t}$ is the stock order imbalance using the trading volume of stock i on day d from time t−1 to t, estimated using Lee and Ready (1991) algorithm. $StockVol_{i,d,t}$ and $OptionVol_{i,d,t}$ are the natural trading volume of stocks i and options underlying stocks i on day d from time t−1 to t, respectively. $Spread_{i,d,t}$ denotes the average 1-minute quote spread of stock i on day d from time t−1 to t. $Turnover_{i,d}$ is the monthly average of daily turnover of stock i at the most recent month. $Volatility_{i,d}$ is the annualized volatility of stock i at the most recent month. $Size_{i,d}$ is the natural logarithm of market capitalization of stock i at the most recent quarter end. $BTM_{i,d}$ is the natural logarithm of equity book to market ratio of stock i at the most recent quarter end. $Ret_{i,d-1}$($Ret_{i,d-20,d-2}$ and $Ret_{i,d-120,d-21}$) is the daily (cumulative) returns of stock i on day d−1 (over day d−20 to d−2 and d−120 to d−21). All continuous independent variables are standardized for ease of interpretation. Columns (1) to (13) show the results for 13 half-hour intraday intervals. Column (14) shows the regression results for thirteen 30-minute intervals altogether (columns 1 to 13). The robust t-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

Table 3. The Stock Return Predictability of OOI by Decompositions

	(1) t = 10:00	(2) t = 10:30	(3) t = 11:00	(4) t = 11:30	(5) t = 12:00	(6) t = 12:30	(7) t = 13:00	(8) t = 13:30	(9) t = 14:00	(10) t = 14:30	(11) t = 15:00	(12) t = 15:30	(13) t = 16:00
Panel A: Put/call OOI													
OOI_Call	**1.096*** **	**1.088*** **	**0.751** **	**0.690***	**1.507*** **	**1.432*** **	**1.429*** **	**0.953*** **	**0.829** **	**1.008*** **	**1.533*** **	**1.540*** **	**2.747*** **
	(3.30)	**(3.16)**	**(2.40)**	**(1.93)**	**(4.45)**	**(3.89)**	**(4.04)**	**(2.65)**	**(2.35)**	**(2.86)**	**(4.15)**	**(4.36)**	**(7.50)**
OOI_Put	**−0.519***	**−0.861*** **	**−0.695** **	**−1.652*** **	**−1.245*** **	**−0.575***	**−1.044*** **	**−1.570*** **	**−0.769** **	**−1.437*** **	**−1.252*** **	**−1.593*** **	**−2.136*** **
	(−1.84)	**(−2.96)**	**(−2.21)**	**(−5.11)**	**(−3.87)**	**(−1.73)**	**(−3.18)**	**(−4.88)**	**(−2.25)**	**(−4.46)**	**(−3.89)**	**(−4.53)**	**(−6.84)**
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	970,163	820,712	786,045	750,584	723,605	697,570	677,865	665,381	664,243	669,809	678,400	711,085	806,914
Adj. R^2	0.1106	0.1171	0.1178	0.1197	0.1192	0.1205	0.1208	0.1223	0.1201	0.1199	0.1192	0.1172	0.1130
Panel B: Open/close OOI													
OOI_Open	**1.245*** **	**1.518*** **	**1.295*** **	**1.514*** **	**2.017*** **	**1.513*** **	**1.901*** **	**1.698*** **	**1.080*** **	**1.921*** **	**2.001*** **	**2.076*** **	**3.662*** **
	(3.70)	**(4.44)**	**(4.15)**	**(4.45)**	**(5.89)**	**(4.24)**	**(5.35)**	**(4.83)**	**(3.00)**	**(5.41)**	**(5.95)**	**(6.03)**	**(9.11)**
OOI_Close	0.629**	0.384	0.039	0.526	0.834***	0.616	0.383	0.529	0.485	0.239	0.552	0.888***	1.082***
	(2.22)	(1.23)	(0.12)	(1.51)	(2.58)	(1.64)	(1.04)	(1.43)	(1.49)	(0.75)	(1.61)	(2.64)	(3.30)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	970,163	820,712	786,045	750,584	723,605	697,570	677,865	665,381	664,243	669,809	678,400	711,085	806,914
Adj. R^2	0.1108	0.1170	0.1177	0.1195	0.1193	0.1206	0.1209	0.1225	0.1202	0.1197	0.1191	0.1170	0.1132
Panel C: Deeper categories													
OOI_Call_Open	**1.032*** **	**1.142*** **	**1.261*** **	**0.756** **	**1.432*** **	**1.507*** **	**1.394*** **	**1.105*** **	**0.665***	**1.135*** **	**1.634*** **	**1.432*** **	**2.783*** **
	(2.90)	**(3.23)**	**(3.86)**	**(2.14)**	**(4.15)**	**(4.13)**	**(3.71)**	**(2.98)**	**(1.73)**	**(3.20)**	**(4.55)**	**(3.92)**	**(7.01)**
OOI_Call_Close	0.650**	0.302	−0.451	0.050	0.776**	0.504	0.558	0.227	0.471	0.167	0.380	0.741**	0.863***
	(2.22)	(0.93)	(−1.40)	(0.14)	(2.38)	(1.31)	(1.50)	(0.64)	(1.43)	(0.50)	(1.07)	(2.20)	(2.84)
OOI_Put_Open	**−0.664** **	**−0.976*** **	**−0.583***	**−1.513*** **	**−1.378*** **	**−0.481**	**−1.329*** **	**−1.443*** **	**−0.829** **	**−1.680*** **	**−1.186*** **	**−1.634*** **	**−2.375*** **
	(−2.22)	**(−3.19)**	**(−1.83)**	**(−4.72)**	**(−4.20)**	**(−1.42)**	**(−3.96)**	**(−4.24)**	**(−2.37)**	**(−4.97)**	**(−3.63)**	**(−4.81)**	**(−7.18)**
OOI_Put_Close	−0.128	−0.277	−0.614**	−0.921**	−0.227	−0.468	−0.001	−0.704**	−0.083	−0.202	−0.414	−0.433	−0.439
	(−0.47)	(−0.90)	(−2.01)	(−2.55)	(−0.69)	(−1.33)	(−0.00)	(−2.11)	(−0.24)	(−0.58)	(−1.23)	(−1.29)	(−1.41)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	970,163	820,712	786,045	750,584	723,605	697,570	677,865	665,381	664,243	669,809	678,400	711,085	806,914
Adj. R^2	0.1110	0.1172	0.1181	0.1198	0.1195	0.1209	0.1210	0.1226	0.1203	0.1200	0.1195	0.1174	0.1132

Note. This table repeats the Fama and MacBeth (1973) regression of OOI predicting subsequent returns in Table 2 and OOI is decomposed into different categories. Panels A and B show the results of call OOI/put OOI and open OOI/close OOI, respectively. Panel C further decomposes the OOI into four deeper categories, which are put open OOI, call open OOI, put close OOI, and put open OOI, respectively. All continuous independent variables are standardized for ease of interpretation. The robust t-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

Table 4. The Stock Return Predictability of OOI by Moneyness

	(1) t = 10:00	(2) t = 10:30	(3) t = 11:00	(4) t = 11:30	(5) t = 12:00	(6) t = 12:30	(7) t = 13:00	(8) t = 13:30	(9) t = 14:00	(10) t = 14:30	(11) t = 15:00	(12) t = 15:30	(13) t = 16:00
Panel A: Out-of-the-Money													
OOI_OTM	**0.841*** **	**0.890*** **	**0.781***	**0.688***	**1.047*** **	**0.885*** **	**0.945*** **	**0.627**	**1.273*** **	**1.587*** **	**1.407*** **	**1.636*** **	**2.092*** **
	(2.73)	**(2.63)**	**(1.91)**	**(1.79)**	**(2.72)**	**(2.18)**	**(2.31)**	**(1.64)**	**(2.97)**	**(4.20)**	**(3.35)**	**(3.72)**	**(5.66)**
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	617,961	522,439	498,049	476,270	458,057	440,086	426,920	418,027	418,990	423,733	431,284	450,979	518,186
Adj. R^2	0.1315	0.1382	0.1387	0.1417	0.1414	0.1404	0.1407	0.1452	0.1421	0.1411	0.1410	0.1378	0.1307
Panel B: At-the-Money													
OOI_ATM	0.730**	0.665**	−0.077	0.863***	1.094***	0.516	0.777**	1.727***	0.011	1.326***	1.563***	1.011**	1.692***
	(2.06)	(2.15)	(−0.21)	(2.69)	(2.89)	(1.36)	(1.99)	(4.57)	(0.03)	(3.54)	(4.03)	(2.39)	(4.54)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	608,030	520,513	493,729	464,802	443,681	424,879	410,168	401,716	399,995	405,720	410,773	437,107	505,886
Adj. R^2	0.1403	0.1487	0.1503	0.1513	0.1529	0.1515	0.1546	0.1545	0.1542	0.1527	0.1524	0.1494	0.1419
Panel C: In-the-Money													
OOI_ITM	1.416***	0.668	1.373***	−0.253	1.081*	1.561***	0.689	1.514**	1.139**	0.816	0.710	0.701	1.790***
	(3.10)	(1.30)	(2.66)	(−0.46)	(1.93)	(2.82)	(1.21)	(2.55)	(1.97)	(1.30)	(1.19)	(1.14)	(2.97)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	363,572	291,607	268,636	250,107	236,133	224,567	215,660	210,275	208,709	211,420	214,956	228,867	272,640
Adj. R^2	0.1455	0.1537	0.1562	0.1597	0.1622	0.1622	0.1675	0.1682	0.1663	0.1651	0.1654	0.1585	0.1481

Note. This table repeats the Fama and MacBeth (1973) regression results of OOI predicting subsequent returns in Table 2 and OOI is decomposed by moneyness categories. Panel A, B, and C show the return predictability of OOI constructed by out-of-the-money, at-the-money, and in-the-money option contracts, respectively. A call (put) option is defined as OTM if the ratio of strike to stock price is between 1.05 and 1.20 (0.80 and 0.95), as ITM if the ratio of strike to stock price is between 0.80 and 0.95 (0.80 and 0.95), and as ATM if the ratio of strike to stock price is between 0.95 and 1.05. All continuous independent variables are standardized for ease of interpretation. The robust t-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

Table 5. The Stock Return Predictability of OOI by Investor Type

	(1) t = 10:00	(2) t = 10:30	(3) t = 11:00	(4) t = 11:30	(5) t = 12:00	(6) t = 12:30	(7) t = 13:00	(8) t = 13:30	(9) t = 14:00	(10) t = 14:30	(11) t = 15:00	(12) t = 15:30	(13) t = 16:00
Panel A: Small Customer													
OOI_Small_Cust	**1.385*****	**1.381*****	**0.889****	**1.538*****	**1.518*****	**1.336*****	**1.289*****	**1.939*****	**1.097*****	**1.509*****	**1.950*****	**1.714*****	**3.419*****
	(3.36)	**(3.33)**	**(2.45)**	**(3.92)**	**(4.19)**	**(3.32)**	**(3.23)**	**(4.79)**	**(2.82)**	**(3.87)**	**(4.97)**	**(4.38)**	**(9.48)**
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	446,039	507,689	536,762	544,753	547,463	543,466	540,317	539,532	544,852	554,848	566,017	595,200	670,850
Adj. R^2	0.1444	0.1397	0.1373	0.1372	0.1358	0.1355	0.1342	0.1346	0.1319	0.1307	0.1297	0.1270	0.1224
Panel B: Middle Customer													
OOI_Mid_Cust	−2.787	16.015	−1.471	8.019	−94.910	−0.132	37.568	11.701	26.839**	−21.610**	−0.768	−5.456	27.071*
	(−0.93)	(1.11)	(−0.17)	(0.65)	(−1.23)	(−0.00)	(0.60)	(0.51)	(2.21)	(−2.10)	(−0.05)	(−0.36)	(1.90)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	35,177	29,363	26,286	24,258	22,886	21,334	20,514	19,926	19,514	19,917	20,468	22,309	28,957
Adj. R^2	0.3554	0.4244	0.4768	0.5058	0.5283	0.5933	0.6177	0.6435	0.6567	0.6339	0.6346	0.5466	0.4072
Panel C: Large Customer													
OOI_Large_Cust	−7.413	87.051	−23.542	−7.112	−23.155	26.640	13.781**	−1.890	3.560	9.432	−4.115	−28.405	−1.490
	(−0.84)	(1.33)	(−1.42)	(−1.01)	(−1.02)	(0.69)	(2.04)	(−0.28)	(0.61)	(0.82)	(−0.38)	(−1.47)	(−0.15)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	16,836	14,681	13,226	12,294	11,109	10,196	9,706	9,188	9,124	9,342	9,739	10,600	13,720
Adj. R^2	0.7660	0.8536	0.9021	0.9384	0.9529	0.9739	0.9779	0.9870	0.9828	0.9799	0.9764	0.9584	0.8614
Panel D: Firm													
OOI_Firm	91.090	−0.810	1.179	0.529	−0.494	0.091	−1.405	−2.250	1.847	−1.428	−0.661	−2.269*	−1.924
	(1.01)	(−0.56)	(0.86)	(0.41)	(−0.42)	(0.07)	(−0.98)	(−1.53)	(1.39)	(−0.99)	(−0.52)	(−1.76)	(−1.55)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	48,161	63,411	68,444	66,398	64,127	61,181	58,559	57,878	59,056	61,310	63,396	68,188	76,497
Adj. R^2	0.2712	0.2703	0.2571	0.2633	0.2722	0.2817	0.2812	0.2772	0.2719	0.2686	0.2682	0.2545	0.2400

Note. This table repeats the Fama and MacBeth (1973) regression results of OOI predicting subsequent returns in Table 2 and OOI is decomposed by investor types. Panels A, B, C, and D show the predictability of OOI using options trades from small customers, middle customers, large customers, and firms, respectively. The customer OOI combines CBOE and ISE options trading volume under the type of small, middle, and large customers (including non-professional and professional customers). Firm OOI includes the options trading volume under the "Firm" type in ISE (including priority account of institutions and brokers/dealers) and options trading volume under the "Firm" and "Broker/Dealer" type in CBOE. Firm OOI represents the informed trading by institutions and brokers/dealers. All continuous independent variables are standardized for ease of interpretation. The robust t-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

Table 6. The Stock Return Predictability of OOI: Opening, Middle, and Closing

Panel A: The next-day stock return predictability			
Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
OOI	1.960***	2.645***	3.605***
	(6.82)	(9.26)	(12.61)
Control	Yes	Yes	Yes
# of Obs.	1,142,658	1,051,026	1,013,032
Adj. R^2	0.0971	0.0988	0.0986
Panel B: Subsample analysis			
Sample period =	Full sample	Pre-sample (2017-2019)	Post-sample (2020-2022)
	(1)	(2)	(3)
OOI_Opening	1.799***	2.506***	1.213*
	(4.05)	(4.23)	(1.88)
OOI_Midday	2.093***	1.782***	2.351***
	(5.28)	(3.62)	(3.93)
OOI_Closing	3.577***	3.119***	3.957***
	(9.47)	(5.81)	(7.53)
Control	Yes	Yes	Yes
# of Obs.	707,499	269,198	438,301
Adj. R^2	0.1188	0.0850	0.1468
Panel C: The long-term stock return predictability			
Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
Next day	1.960***	2.645***	3.605***
	(6.82)	(9.26)	(12.61)
1 week	0.556***	0.861***	1.183***
	(4.31)	(7.12)	(8.89)
2 weeks	0.395***	0.539***	0.659***
	(4.14)	(5.39)	(6.80)
3 weeks	0.288***	0.393***	0.362***
	(3.76)	(4.87)	(4.34)
4 weeks	0.273***	0.338***	0.303***
	(4.10)	(5.01)	(4.25)
Panel D: The out-of-sample analyses			
Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
R^2_{OOS}	0.381	0.433	0.452
Diebold-Mariano test	$DM_{12} = -0.56$	$DM_{23} = 2.04$**	$DM_{13} = 2.72$***

Note. This table shows the Fama and MacBeth (1973) regression of OOI predicting subsequent returns under coarser time intervals. Opening, midday, and closing intervals consist of trading over [9:30,11:30), [11:30,14:00), and [14:00,16:00), respectively. Panel A shows the predictive power of OOI over three intraday intervals separately. Panel B shows the predictive power of OOI over three intraday intervals together and for two subsample periods: pre-sample (July 2017 to December 2019) and post-sample (January 2020 to December 2022). Panel C examines the predictive power of OOI on long-term stock returns up to next four weeks. The long-run stock returns are defined as the average daily stock returns over corresponding horizons. Panel D shows the out-of-sample R^2 and Diebold-Mariano test statistics between two corresponding intervals. The robust *t*-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

Table 7. Options and Other Informed Traders

Panel A: Institutional investors Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
OOI×TRA	**1.109****	**0.871***	**1.190****
	(2.38)	**(1.71)**	**(2.28)**
OOI×DED	**−0.778**	**−1.143****	**0.576**
	(−1.45)	**(−2.02)**	**(1.02)**
OOI×QIX	**−0.043**	**−0.719**	**−1.553*****
	(−0.08)	**(−1.35)**	**(−2.65)**
OOI	1.799***	3.156***	3.456***
	(3.40)	(5.60)	(6.07)
TRA	4.005***	4.407***	4.362***
	(4.73)	(5.01)	(5.02)
DED	1.453**	1.032	1.171
	(2.03)	(1.42)	(1.55)
QIX	1.891**	1.672**	1.838**
	(2.55)	(2.30)	(2.41)
Control	Yes	Yes	Yes
# of Obs.	1,142,658	1,051,026	1,013,032
Adj. R^2	0.1000	0.1018	0.1018
Panel B: Short sellers Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
OOI×HighBC	**0.719**	**0.850**	**1.272****
	(1.49)	**(1.58)**	**(2.43)**
OOI	1.553***	2.186***	2.947***
	(4.27)	(5.42)	(7.43)
HighBC	1.268**	1.774***	1.411*
	(2.00)	(2.77)	(1.95)
Control	Yes	Yes	Yes
# of Obs.	1,142,658	1,051,026	1,013,032
Adj. R^2	0.0976	0.0994	0.0992
Panel C: Insiders Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
OOI×Insider	**−2.844**	**8.366**	**7.392**
	(−0.86)	**(0.74)**	**(1.35)**
OOI	1.934***	2.652***	3.591***
	(6.65)	(9.27)	(12.45)
Insider	−1.020	−3.024	8.068
	(−0.40)	(−1.05)	(1.18)
Control	Yes	Yes	Yes
# of Obs.	1,142,658	1,051,026	1,013,032
Adj. R^2	0.0972	0.0988	0.0985

Note. This table examines the effect of informed trader intensity on the stock return predictability of OOI. In Panel A, *TRA*, *DED*, *QIX* are indicators that equals to one if the institutional ownership by transient, dedicated, and quasi-indexer institutions is higher than the quarterly cross-sectional median, respectively. The classification is based on Bushee (1998). Institutional ownership is defined as the sum of shares held by each institution divided by the total shares outstanding. In Panel B, *HighBC* is an indicator that equals one if a firm's daily borrower concentration is above the cross-sectional median, and zero otherwise. In Panel C, *Insider* is an indicator that equals one if a day involves insider options transactions. All continuous independent variables are standardized for ease of interpretation. The robust *t*-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

Table 8. Liquidity Timing

Panel A: Options expiration dates			
Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
OOI*×*TripleWitching	**0.052**	**0.014**	**0.085****
	(1.32)	**(0.37)**	**(2.31)**
OOI*×3*rdFriday	**0.040**	**0.043**	**0.092****
	(0.91)	**(0.97)**	**(2.08)**
OOI	1.868***	2.588***	3.428***
	(6.70)	(9.10)	(12.00)
TripleWitching	−0.639	−0.672	−0.445
	(−1.40)	(−1.34)	(−1.05)
3*rdFriday*	−0.141	−0.454	−0.188
	(−0.23)	(−0.79)	(−0.31)
Control	Yes	Yes	Yes
# of Obs.	1,142,658	1,051,026	1,013,032
Adj. R^2	0.0925	0.0937	0.0934
Panel B: Option v.s. stock liquidity condition			
Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
OOI*×*OptionVol	**1.476*****	**2.049*****	**2.723*****
	(4.10)	**(5.81)**	**(7.98)**
OOI*×*StockVol	**−0.589***	**−0.807*****	**−1.423*****
	(−1.66)	**(−2.60)**	**(−4.27)**
OOI	2.535***	3.501***	4.929***
	(7.24)	(9.51)	(13.85)
OptionVol	−0.304	−0.573	−0.779*
	(−0.76)	(−1.42)	(−1.84)
StockVol	−1.886**	−1.949**	−2.181**
	(−2.05)	(−2.11)	(−2.41)
Control	Yes	Yes	Yes
# of Obs.	1,142,658	1,051,026	1,013,032
Adj. R^2	0.0977	0.0990	0.0990
Panel C: Option relative liquidity condition			
Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
OOI*×*RelativeSpread	**−1.225****	**−0.743****	**−0.898*****
	(−2.45)	**(−2.19)**	**(−3.10)**
OOI	1.616***	2.536***	3.598***
	(4.48)	(8.74)	(12.70)
RelativeSpread	−0.573*	−0.640*	−0.427
	(−1.73)	(−1.79)	(−1.22)
Control	Yes	Yes	Yes
# of Obs.	1,142,298	1,050,681	1,012,695
Adj. R^2	0.0967	0.0983	0.0982

Note. This table examines the liquidity timing hypothesis in option market. The dependent variable is next-day stock returns. In Panel A, 3*rdFriday* is an indicator that equals to one if a trading day is the third Friday of the month, and zero otherwise. *TripleWitching* is an indicator that equals to one if a trading day is the third Friday of March, June, September, or December, and zero otherwise. In Panel B, *OptionVol* and *StockVol* denote the (log) options trading volume and stock trading volume at the corresponding time interval, respectively. In Panel C, *RelativeSpread* denotes the ratio of options bid-ask spread to stocks bid-ask spread. All continuous independent variables are standardized for ease of interpretation. The robust *t*-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

Table 9. Private and Processed Public Information Channel

Panel A: *Future* news predicts options order imbalance			
Dependent variable =	*OOI*		
Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
POS_NSCH	**0.002**	**−0.006**	**0.013*** **
	(0.44)	**(−1.39)**	**(3.20)**
Control	Yes	Yes	Yes
# of Obs.	1,145,903	1,053,905	1,015,713
Adj. R^2	0.0145	0.0129	0.0132
Panel B: Options order imbalance predicts *future* news			
Dependent variable =	$POS_NSCH_{i,d+1}$		
Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
OOI	**0.006**	**−0.011**	**0.045*** **
	(0.37)	**(−0.75)**	**(2.85)**
Control	Yes	Yes	Yes
# of Obs.	1,145,903	1,053,905	1,015,713
Adj. R^2	0.0067	0.0063	0.0066
Panel C: Stock returns predictability of OOI conditional on *future* news			
Dependent variable =	$Ret_{i,d+1}$		
Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
OOI×POS_NSCH	**3.340**	**5.215**	**8.881***
	(0.62)	**(1.52)**	**(1.89)**
OOI	1.901***	2.546***	3.486***
	(6.47)	(8.75)	(12.33)
POS_NSCH	81.105***	78.901***	79.636***
	(15.18)	(26.54)	(25.52)
Control	Yes	Yes	Yes
# of Obs.	1,142,658	1,051,026	1,013,032
Adj. R^2	0.1022	0.1043	0.1042
Panel D: Stock returns predictability of OOI conditional *public* news			
Dependent variable =	$Ret_{i,d+1}$		
Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
OOI×POS_NSCH	**4.611***	**0.334**	**−0.466**
	(1.94)	**(0.14)**	**(−0.15)**
OOI	1.929***	2.578***	3.571***
	(6.73)	(8.94)	(12.20)
POS_NSCH	1.015	−0.821	0.371
	(0.48)	(−0.44)	(0.14)
Control	Yes	Yes	Yes
# of Obs.	1,142,658	1,051,026	1,013,032
Adj. R^2	0.0973	0.1005	0.0989

Note. This table examines the effect of firm-specific news on the stock return predictability of OOI. Panel A examines how OOI responds to future positive non-scheduled news. *POS_NSCH* is an indicator that equals one if at least one positive non-scheduled earnings or analyst-related news will be released after today's stock market close and before next day's market open. Panel B examines whether OOI predicts the release of future positive non-scheduled news. The coefficient of OOI is multiplied by 100 for presentation purpose. Panel C examines the effect of private information on stock return predictability of OOI. In Panel D, *POS_NSCH* is an indicator that equals one if at least one positive non-scheduled firm news was released during the overnight before today's stock market open. All continuous independent variables are standardized for ease of interpretation. The robust *t*-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

Table 10. Macroeconomic Information

Panel A: FOMC meetings Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
***OOI*×*FOMC*$^{-1}$**	**0.146***** **(2.74)**	**0.078** **(1.61)**	**0.203***** **(3.24)**
***OOI*×*FOMC*^{0}**	**0.093*** **(1.75)**	**0.122***** **(2.70)**	**0.158***** **(2.86)**
OOI*×*Press	**0.122**** **(2.30)**	**0.094*** **(1.79)**	**0.060** **(1.32)**
OOI	1.621*** (5.90)	2.347*** (8.57)	3.177*** (11.48)
FOMC^{-1}	0.141 (0.33)	−0.060 (−0.14)	−0.245 (−0.56)
FOMC^{0}	0.105 (0.19)	0.252 (0.40)	−0.029 (−0.05)
Press	−1.742** (−2.29)	−1.664** (−2.21)	−1.219* (−1.91)
Control	Yes	Yes	Yes
# of Obs.	1,137,255	1,046,040	1,008,388
Adj. R^2	0.0903	0.0914	0.0909
Panel B: VIX shock Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
***OOI*×|Δ*VIX*|**	**0.165***** **(2.76)**	**0.139**** **(2.07)**	**0.185***** **(2.82)**
OOI	1.610*** (5.84)	2.432*** (9.36)	3.260*** (11.82)
|Δ*VIX*|	−0.565 (−0.66)	−0.794 (−0.92)	−0.687 (−0.80)
Control	Yes	Yes	Yes
# of Obs.	1,137,255	1,046,040	1,008,388
Adj. R^2	0.0949	0.0964	0.0962
Panel C: Interest rate shock Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
***OOI*×|Δ*IR*|**	**0.059** **(1.02)**	**0.120**** **(2.27)**	**0.223***** **(3.96)**
OOI	1.831*** (6.54)	2.378*** (8.62)	3.215*** (11.49)
|Δ*IR*|	−1.086 (−1.26)	−1.059 (−1.21)	−1.018 (−1.20)
Control	Yes	Yes	Yes
# of Obs.	1,133,490	1,042,609	1,005,143
Adj. R^2	0.0950	0.0965	0.0962

Note. This table examines the effect of macroeconomic events on stock return predictability of OOI. In Panel A, *FOMC*$^{-1}$, *FOMC*^{0}, and *Press* is respectively an indicator that equals to one if a day is the previous day, the first day of an FOMC meeting, and with a press conference, otherwise zero. In Panel B, |Δ*VIX*| is the absolute daily change in VIX. In Panel C, |Δ*IR*| is the absolute daily change in 1-year nominal zero coupon bond yield. All continuous independent variables are standardized for ease of interpretation. The robust *t*-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

Online Appendices for

When Does Intraday Options Trading Predict Stock Returns?

Online Appendix A. Variable Definitions

Variables	Definition
$\lvert\Delta IR_d\rvert$	The absolute daily change in 1-year nominal zero coupon bond yield.
$\lvert\Delta VIX_d\rvert$	The absolute daily change in VIX.
$3rdFriday_d$	An indicator equals to one if day d is the third Friday of the month, and zero otherwise.
$BorrowFee_{i,d}$	The indicative stock borrow fee of stock i on day d, collected from Markit. It is expressed in percentage points.
$BTM_{i,d}$	The natural logarithm of equity book to market ratio of stock i at the most recent quarter end.
$DED_{i,d}$	An indicator that equals one if the dedicated institutional ownership of stock i is higher than the quarterly cross-sectional median, and zero otherwise. Dedicated institutional ownership is defined as the sum of shares held by each dedicated institution divided by the total shares outstanding. The classification is based on Bushee (1998). Dedicated investors have a high concentration of ownership, low turnover, and near-zero momentum.
$FOMC_d^0$	An indicator equals to one if day d is the first day of an FOMC meeting, and zero otherwise.
$FOMC_d^{-1}$	An indicator equals to one if day d is the day prior to an FOMC meeting, and zero otherwise.
$HighBC_{i,d}$	An indicator equals to one if stock i's borrower concentration on day d is above the cross-sectional median, and zero otherwise.
$Insider_{i,d}$	An indicator equals to one if day d involves insider options transactions of stock i, and zero otherwise.
$JumpSize_{i,d+1}$	Stock jump size of stock i on day d+1, estimated based on Aït-Sahalia, Li, and Li (2025).
$OBPC_{i,d,t}$	Put-call ratio underlying stock i on day d+1, using options buying to open new positions only, estimated based on Pan and Poteshman (2006).
$OOI_{i,d,t}$	Options order imbalance of underlying stock i at time interval t of day d. It is estimated as (B−S) / (B+S), where B (S) is the aggregate buyer-initiated (seller-initiated) trading volume. Buyer-initiated (seller-initiated) trading volume consists of open buy and close buy (open sell and close sell) options volume.
OOI_ATM	Options order imbalance using at-the-money (ATM) options only. A call (put) option is defined as ATM if the strike to stock price ratio is between 0.95 and 1.05.
OOI_Call	Options order imbalance using call options only.
OOI_Call_Close	Options order imbalance using call options closing existing positions only.
OOI_Call_Open	Options order imbalance using call options open new positions only.
OOI_CBOE	Options order imbalance using CBOE open/close data only.
OOI_Close	Options order imbalance using options closing existing positions only.
OOI_Firm	Options order imbalance using options traded by "firm." In CBOE, we combine "firm" (*clearing* firm of options clearing corporation (OCC), *or* by a non-trading permit holder affiliate) and "broker/dealer" (broker or dealer [including foreign brokers/dealers], *or* by a joint back office

account). In ISE, we directly use the "firm" type (an institution like Morgan Stanley or Goldman enters a trade for their own account. ISE FIRM trades can be PROPRIETARY trades, executed on behalf of their own trading account or for another BROKER/DEALER who is not a member of the exchange).

OOI_ISE Options order imbalance using ISE open/close data only.

OOI_ITM Options order imbalance using in-the-money (ITM) options only. A call (put) option is defined as ITM if the ratio of strike to stock price is between 0.80 and 0.95 (0.80 and 0.95).

OOI_Large_Cust Options order imbalance using options traded by public customers *and* professional customers with trade a trading volume larger or equal to 200 contracts each.

OOI_Mid_Cust Options order imbalance using options traded by public customers *and* professional customers with a trade volume of 100-199 contracts each.

OOI_Open Options order imbalance using options opening new positions only.

OOI_OTM Options order imbalance using out-of-the-money (OTM) options only. A call (put) option is defined as OTM if the ratio of strike to stock price is between 1.05 and 1.20 (0.80 and 0.95).

OOI_Put Options order imbalance using put options only.

OOI_Put_Close Options order imbalance using put options closing existing positions only.

OOI_Put_Open Options order imbalance using put options open new positions only.

OOI_Small_Cust Options order imbalance using options traded by public customers *and* professional customers with trade volume less than 100 contracts each. Professional customers are defined as a person or entity that (i) is not a broker/dealer in securities, and (ii) places more than 390 orders in listed options per day on average during a calendar month for their beneficial account(s).

$OptionVol_{i,d,t}$ The natural logarithm of options trade volume of underlying stock i at time interval t of day d.

$POS_NSCH_{i,d}$ An indicator equals one if there is at least one positive non-scheduled earnings- or analyst-related news released during the overnight after (before) day d's stock market close (open) for private (processed public) information channel, and zero otherwise.

$Press_d$ An indicator equals to one if day d is the second day of an FOMC meeting with a press conference, and zero otherwise.

$QIX_{i,d}$ An indicator that equals one if the quasi-indexer institutional ownership of stock i is higher than the quarterly cross-sectional median, and zero otherwise. Quasi-indexer institutional ownership is defined as the sum of shares held by each quasi-indexer institution divided by the total shares outstanding. The classification is based on Bushee (1998). Quasi-indexers have high diversification, low turnover, and low momentum.

$RelativeSpread_{i,d}$ The relative bid-ask spreads of options relative to underlying stocks. The bid-ask spread of options is defined as national best (lowest) ask minus national best (highest) bid, scaled by the national best quote midpoint. We average bid-ask spreads across eligible option contracts to obtain stock-day level variable. The bid-ask spread of stocks is defined as

	national best (lowest) ask minus national best (highest) bid, scaled by the closing price. It is scaled by 10,000.
$Ret_{i,d+1}$	The daily returns of stock i on day d+1. It is calculated using the last NBBO quote midpoints before 4 p.m. to avoid bid-ask bounce. It is expressed in basis points.
$Ret_{i,d-1}$	The daily returns of stock i on day d−1.
$Ret_{i,d-120,d-21}$	The cumulative returns of stock i over day d−120 to d−21.
$Ret_{i,d-20,d-2}$	The cumulative returns of stock i over day d−20 to d−2.
$RSkew_{i,d+1}$	The realized skewness of stock i on day d+1, estimated based on Amaya et al. (2015) and expressed in basis points.
$Size_{i,d}$	The natural logarithm of market capitalization of stock i at the most recent quarter end.
$SOI_{i,d,t}$	Stock order imbalance of stock i at time interval t of day d. It is estimated by Lee and Ready (1991) algorithm.
$Spread_{i,d,t}$	Average 1-minute stock bid-ask spread of stock i over time interval t of day d.
$StockVol_{i,d,t}$	The natural logarithm of stock trade volume of stock i at time interval t of day d.
$TRA_{i,d}$	An indicator that equals one if the transient institutional ownership of stock i is higher than the quarterly cross-sectional median, and zero otherwise. Transient institutional ownership is defined as the sum of shares held by each transient institution divided by the total shares outstanding. The classification is based on Bushee (1998). Transient investors have the highest trading turnover, the highest use of momentum strategies, and relatively high diversification.
$TripleWitching_{d}$	An indicator equals to one if day d is the third Friday of March, June, September, or December, and zero otherwise.
$Turnover_{i,d}$	The monthly average of daily turnover of stock i at the most recent month. The daily turnover is defined as the trading volume scaled by the number of total shares outstanding.
$Volatility_{i,d}$	The annualized volatility of stock i at the most recent month end. The volatility is estimated using daily stock returns every month.

Online Appendix B. Additional Figures and Tables

Figure OA1. Intraday average of additional variables

Figure OA2. Intraday OOI coefficient of Predicting Rest-of-Day Stock Returns

Figure OA3. Intraday OOI coefficient of Predicting the Next Half-an-Hour Stock Returns

Figure OA4. Subsample Analysis

Table OA1. Descriptive Statistics of Additional Variables

Table OA2. OOI Predicts Rest-of-Day and Overnight Stock Returns Predictability

Table OA3. OOI Predicts Half-Hour and Overnight Stock Returns

Table OA4. Horse-Racing OOI in Predicting Next-Day Stock Returns

Table OA5. Additional Analyses

Table OA6. Robustness Checks

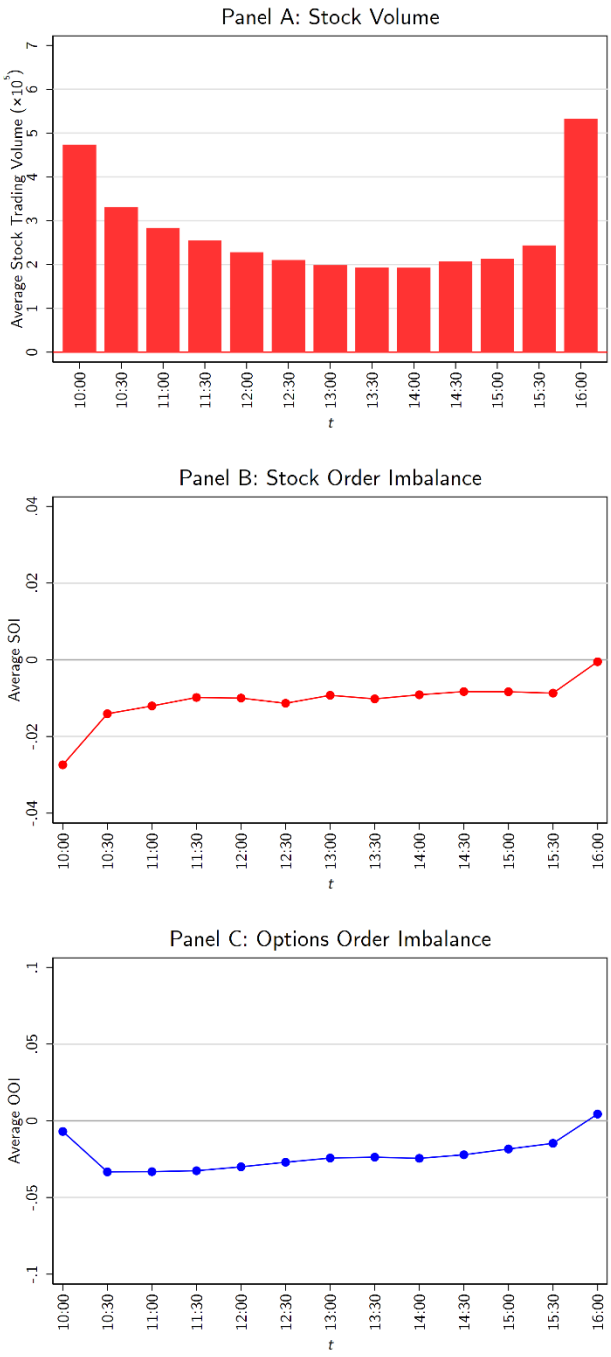


Figure OA1. **Intraday average of additional variables.**

This figure shows the average variables across every 30-minute interval. Panels A, B, and C show that of stock volume, stock order imbalance and options order imbalance, respectively. The sample period spans from July 2017 to December 2022.

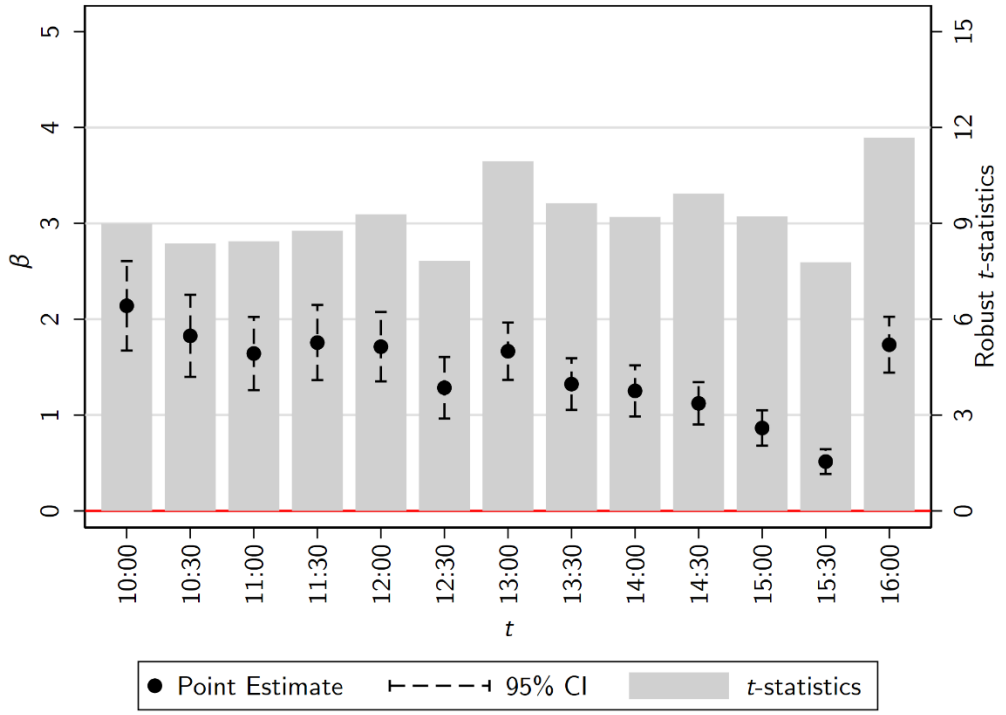


Figure OA2. **Intraday OOI coefficient of Predicting Rest-of-Day Stock Returns.**
It shows the $\hat{\beta}$ with a 95% confidence interval, and corresponding robust *t*-statistics shown in Table OA2 with standardization. The sample period is between July 2017 and December 2022.

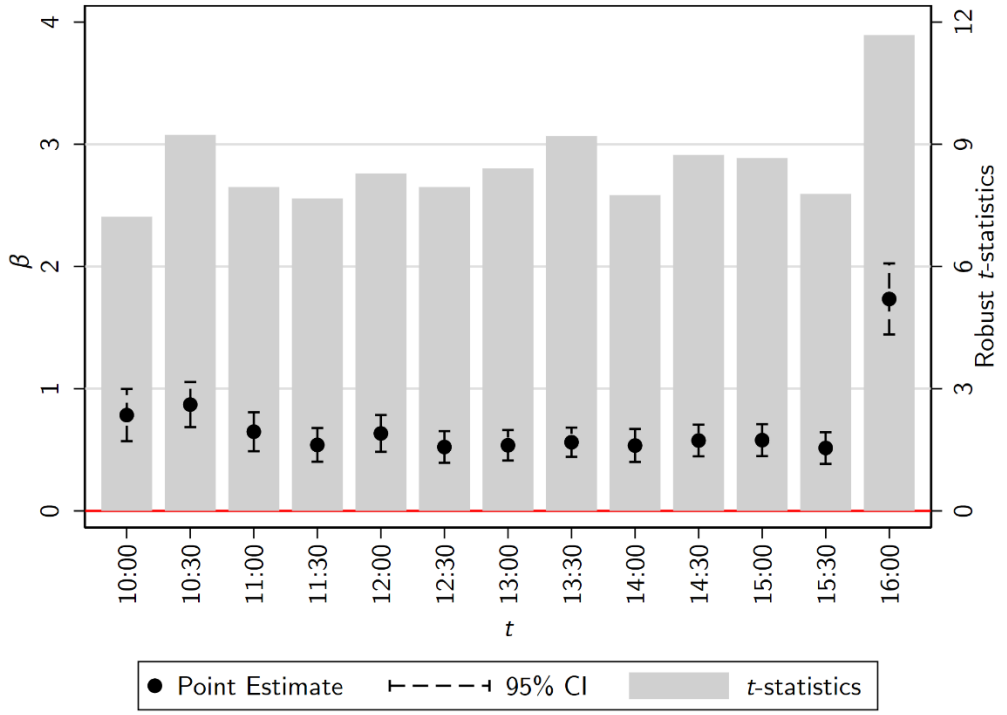


Figure OA3. **Intraday OOI coefficient of Predicting the Next Half-an-Hour Stock Returns.** It shows the $\hat{\beta}$ with a 95% confidence interval, and corresponding robust t-statistics shown in Table OA3 with standardization. The sample period is between July 2017 and December 2022.

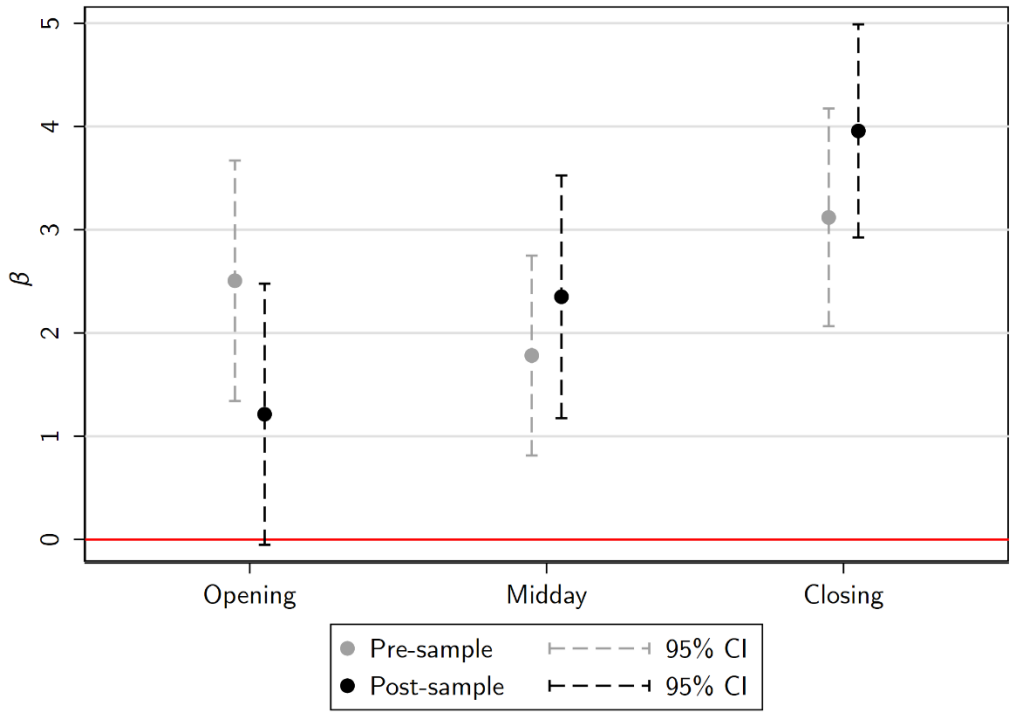


Figure OA4. Subsample Analysis.

This figure shows the $\hat{\beta}$ with 95% confidence interval for the pre-sample period (July 2017-December 2019) and the post-sample period (January 2020-December 2022), respectively. The full sample period is between July 2017 and December 2022.

Table OA1. Descriptive Statistics of Additional Variables

	N	Mean	*SD*	Min	Q1	Median	Q3	Max
OOI_Call	1,340,350	−0.052	0.550	−1.000	−0.333	0.000	0.143	1.000
OOI_Put	1,340,350	−0.044	0.357	−1.000	−0.053	0.000	0.000	1.000
OOI_Open	1,340,350	0.041	0.577	−1.000	−0.256	0.000	0.400	1.000
OOI_Close	1,340,350	−0.050	0.358	−1.000	−0.091	0.000	0.000	1.000
OOI_Call_Open	1,340,350	−0.011	0.485	−1.000	−0.172	0.000	0.133	1.000
OOI_Call_Close	1,340,350	−0.041	0.306	−1.000	−0.032	0.000	0.000	1.000
OOI_Put_Open	1,340,350	−0.052	0.318	−1.000	−0.053	0.000	0.000	1.000
OOI_Put_Close	1,340,350	0.008	0.179	−0.900	0.000	0.000	0.000	0.875
$TRA_{i,d}$	1,908,260	0.502	0.500	0.000	0.000	1.000	1.000	1.000
$DED_{i,d}$	1,908,260	0.502	0.500	0.000	0.000	1.000	1.000	1.000
$QIX_{i,d}$	1,908,260	0.502	0.500	0.000	0.000	1.000	1.000	1.000
$BC_{i,d}$	1,908,260	0.500	0.500	0.000	0.000	0.000	1.000	1.000
$Insider_{i,d}$	1,908,260	0.023	0.149	0.000	0.000	0.000	0.000	1.000
$TripleWitchings_{d}$	1,908,260	0.017	0.129	0.000	0.000	0.000	0.000	1.000
$3rdFriday_{d}$	1,908,260	0.050	0.219	0.000	0.000	0.000	0.000	1.000
$RelativeSpread_{i,d}$	1,907,402	0.091	0.141	−0.520	0.022	0.047	0.103	6.821
$POS_NSCH_{i,d}$	1,908,260	0.027	0.163	0.000	0.000	0.000	0.000	1.000
$FOMC_{-1,d}$	1,899,502	0.035	0.183	0.000	0.000	0.000	0.000	1.000
$FOMC_{0,d}$	1,899,502	0.033	0.178	0.000	0.000	0.000	0.000	1.000
$Press_{d}$	1,899,502	0.028	0.166	0.000	0.000	0.000	0.000	1.000
$\lvert\Delta VIX_{d}\rvert$	1,899,502	1.283	1.757	0.000	0.340	0.810	1.580	24.860
$\lvert\Delta IR_{d}\rvert$	1,893,371	0.019	0.027	0.000	0.004	0.009	0.022	0.299
$OBPC_{i,d,t}$	929,813	0.328	0.389	0.000	0.000	0.103	0.667	1.000
$JumpSize_{i,d+1}$	1,596,748	331.095	133.423	−1809.284	244.870	328.531	400.663	1816.198
$RSkew_{i,d+1}$	1,733,861	−0.732	23.262	−50.136	−18.832	−1.290	17.285	50.768

Note. This table shows the descriptive statistics of unstandardized values for additional variables used in the empirical specifications. When the frequency of a variable is intraday, we show the statistics near the market close from 14:00 to 16:00 as references. The sample period spans from July 2017 to December 2022. All unbounded and continuous variables are winsorized at the top and bottom 1% percentile. All variables are defined in Online Appendix A.

Table OA2. OOI Predicts Rest-of-Day and Overnight Stock Returns Predictability

	(1) t = 10:00	(2) t = 10:30	(3) t = 11:00	(4) t = 11:30	(5) t = 12:00	(6) t = 12:30	(7) t = 13:00	(8) t = 13:30	(9) t = 14:00	(10) t = 14:30	(11) t = 15:00	(12) t = 15:30	(13) t = 16:00
OOI	**2.140*** **	**1.826*** **	**1.642*** **	**1.756*** **	**1.713*** **	**1.285*** **	**1.666*** **	**1.323*** **	**1.252*** **	**1.122*** **	**0.865*** **	**0.515*** **	**1.734*** **
	(8.99)	**(8.37)**	**(8.44)**	**(8.77)**	**(9.28)**	**(7.83)**	**(10.94)**	**(9.63)**	**(9.20)**	**(9.93)**	**(9.22)**	**(7.78)**	**(11.68)**
SOI	0.827***	0.659***	0.239	0.544**	0.399*	0.315	0.228	0.512***	0.323**	0.336**	0.429***	0.236***	−2.005***
	(2.91)	(2.65)	(0.96)	(2.55)	(1.91)	(1.64)	(1.24)	(2.81)	(2.04)	(2.22)	(3.28)	(2.64)	(−10.85)
StockVol	−0.745**	−0.134	0.169	0.235	−0.158	−0.090	0.324	0.221	0.287	0.267*	0.175	0.308***	0.823***
	(−2.14)	(−0.40)	(0.59)	(0.91)	(−0.59)	(−0.41)	(1.58)	(1.27)	(1.43)	(1.69)	(1.38)	(2.98)	(3.72)
OptionVol	−1.552**	−1.718***	−1.294**	−0.952*	−0.968**	−1.126**	−1.520***	−0.501	−0.483	0.000	−0.064	0.193	0.170
	(−2.38)	(−3.06)	(−2.41)	(−1.89)	(−2.08)	(−2.41)	(−3.85)	(−1.44)	(−1.50)	(0.00)	(−0.25)	(1.18)	(0.32)
Spread	−0.668*	−0.548	−0.498	−0.390	−0.331	−0.361	−0.520**	−0.137	−0.251	0.053	−0.092	0.164	0.076
	(−1.68)	(−1.61)	(−1.51)	(−1.35)	(−1.24)	(−1.36)	(−2.29)	(−0.67)	(−1.32)	(0.31)	(−0.59)	(1.63)	(0.31)
Turnover	1.313*	1.495**	0.895	−0.080	−0.090	0.052	−0.114	−0.332	0.106	−0.054	−0.389	−0.065	−3.395***
	(1.87)	(2.14)	(1.46)	(−0.14)	(−0.16)	(0.10)	(−0.23)	(−0.74)	(0.29)	(−0.16)	(−1.19)	(−0.29)	(−7.50)
Volatility	−2.419***	−1.862**	−1.502**	−1.014	−1.157*	0.166	−0.302	0.096	0.589	0.177	0.103	0.635***	1.393**
	(−2.82)	(−2.27)	(−2.12)	(−1.48)	(−1.87)	(0.30)	(−0.57)	(0.20)	(1.39)	(0.46)	(0.34)	(3.04)	(2.58)
Size	3.153***	2.609***	1.431*	0.756	1.364*	1.549**	1.103*	0.172	0.758	0.501	0.803**	−0.072	−2.970***
	(3.41)	(2.90)	(1.72)	(1.02)	(1.90)	(2.28)	(1.91)	(0.33)	(1.60)	(1.15)	(2.13)	(−0.27)	(−4.38)
BTM	−0.221	−0.106	−0.194	−0.602	−0.793**	−0.377	−0.688**	−0.643*	−0.426	−0.492*	0.004	−0.460***	−0.025
	(−0.37)	(−0.20)	(−0.40)	(−1.36)	(−2.02)	(−1.01)	(−1.98)	(−1.88)	(−1.40)	(−1.89)	(0.02)	(−3.17)	(−0.07)
Ret_{d-1}	−2.769***	−0.677	−0.038	−0.466	−0.039	0.628	0.876**	1.254***	1.402***	1.145***	1.151***	0.728***	−0.177
	(−3.93)	(−1.05)	(−0.07)	(−0.86)	(−0.07)	(1.38)	(1.97)	(3.08)	(4.19)	(3.66)	(4.29)	(3.92)	(−0.44)
$Ret_{d-20,d-2}$	−1.400*	−0.673	−0.345	−0.361	−0.522	−0.360	−0.290	−0.144	−0.111	0.334	0.524**	0.584***	−0.618
	(−1.88)	(−1.02)	(−0.60)	(−0.70)	(−1.16)	(−0.78)	(−0.67)	(−0.36)	(−0.32)	(1.11)	(2.19)	(3.55)	(−1.35)
$Ret_{d-120,d-21}$	0.050	−0.090	0.284	0.414	0.319	0.510	0.446	0.365	0.065	0.261	0.402*	0.359*	2.212***
	(0.06)	(−0.12)	(0.45)	(0.74)	(0.61)	(1.03)	(0.94)	(0.88)	(0.17)	(0.85)	(1.68)	(1.92)	(3.91)
# of Obs.	971,603	821,884	787,107	751,588	724,531	698,525	678,739	666,378	665,212	670,744	679,417	712,077	808,157
Adj. R^2	0.0937	0.0970	0.0918	0.0896	0.0905	0.0897	0.0905	0.0877	0.0854	0.0822	0.0763	0.0645	0.0934

Note. This table shows the Fama and MacBeth (1973) regression results of OOI predicting rest-of-day returns. Columns (1) to (12) show the results for 12 half-hour intraday intervals: $Ret_{i,d,[t,16:00)} = \alpha + \beta \times OOI_{i,d,t} + \psi' X_{i,d,t} + \epsilon_{i,d,[t,16:00)}$, and column (13) shows the result for the following equation for the last half-hour interval: $Ret_{i,d,\mathrm{ON}} = \alpha + \beta \times OOI_{i,d,t} + \psi' X_{i,d,t} + \epsilon_{i,d,\mathrm{ON}}$, where $Ret_{i,d,[t,16:00)}$ denotes the returns of stock i on day d, estimated by the log difference of the closing price and the NBBO quote midpoints at time t on day d, expressed in basis points. $Ret_{i,d,\mathrm{ON}}$ denotes the overnight return of stock i, estimated by the log difference between the first NBBO quote midpoints after 9:30 on day d+1 and the closing price on day d, expressed in basis points. $OOI_{i,d,t}$ is the options order imbalance using the trading volume of options underlying stock i on day d from time $t-1$ to t. $X_{i,d,t}$ denotes a battery of control variables: $SOI_{i,d,t}$ is the stock order imbalance using the trading volume of stock i on day d from time $t-1$ to t. $StockVol_{i,d,t}$ and $OptionVol_{i,d,t}$ are the natural trading volume of stocks i and options underlying stocks i on day d from time $t-1$ to t, respectively. $Spread_{i,d,t}$ denotes the average 1-minute quote spread of stock i on day d from time $t-1$ to t. $Turnover_{i,d}$ is the monthly average of daily turnover of stock i at the most recent month. $Volatility_{i,d}$ is the annualized volatility of stock i at the most recent month. $Size_{i,d}$ is the natural logarithm of market capitalization of stock i at the most recent quarter end. $BTM_{i,d}$ is the natural logarithm of equity book to market ratio of stock i at the most recent quarter end. $Ret_{i,d\text{-}1}$ ($Ret_{i,d\text{-}20,d\text{-}2}$ and $Ret_{i,d\text{-}120,d\text{-}21}$) is the daily (cumulative) returns of stock i on day d-1 (over day $d-20$ to $d-2$ and $d-120$ to $d-21$). All continuous independent variables are standardized by deducting their mean and then scaled by the standard deviation for ease of interpretation. The robust t-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

Table OA3. OOI Predicts Half-Hour and Overnight Stock Returns

	(1) t = 10:00	(2) t = 10:30	(3) t = 11:00	(4) t = 11:30	(5) t = 12:00	(6) t = 12:30	(7) t = 13:00	(8) t = 13:30	(9) t = 14:00	(10) t = 14:30	(11) t = 15:00	(12) t = 15:30	(13) t = 16:00
OOI	**0.784*** **	**0.870*** **	**0.648*** **	**0.540*** **	**0.634*** **	**0.523*** **	**0.537*** **	**0.562*** **	**0.535*** **	**0.576*** **	**0.579*** **	**0.515*** **	**1.734*** **
	(7.22)	**(9.23)**	**(7.95)**	**(7.67)**	**(8.28)**	**(7.95)**	**(8.41)**	**(9.20)**	**(7.75)**	**(8.74)**	**(8.66)**	**(7.78)**	**(11.68)**
SOI	0.436***	0.038	−0.015	0.022	−0.005	0.132*	0.068	−0.007	−0.109	0.108	0.073	0.236***	−2.005***
	(3.60)	(0.36)	(−0.16)	(0.26)	(−0.07)	(1.67)	(0.87)	(−0.08)	(−1.40)	(1.32)	(0.92)	(2.64)	(−10.85)
StockVol	−0.341**	−0.265**	−0.041	0.165	−0.092	−0.079	−0.091	−0.014	0.041	0.003	−0.126	0.308***	0.823***
	(−2.14)	(−2.10)	(−0.35)	(1.60)	(−1.03)	(−0.90)	(−1.08)	(−0.17)	(0.51)	(0.04)	(−1.55)	(2.98)	(3.72)
OptionVol	−0.613*	−0.451*	−0.247	0.016	−0.230	−0.004	−0.829***	−0.626***	−0.423***	−0.010	−0.530***	0.193	0.170
	(−1.86)	(−1.79)	(−1.10)	(0.09)	(−1.16)	(−0.02)	(−5.02)	(−3.78)	(−2.70)	(−0.06)	(−3.35)	(1.18)	(0.32)
Spread	−0.485***	−0.024	−0.160	−0.079	−0.284***	0.086	−0.284***	−0.311***	−0.252***	0.149	−0.366***	0.164	0.076
	(−2.62)	(−0.17)	(−1.17)	(−0.62)	(−2.73)	(0.76)	(−2.80)	(−3.22)	(−2.62)	(1.62)	(−3.79)	(1.63)	(0.31)
Turnover	0.002	0.218	0.230	0.093	−0.091	0.050	0.084	−0.095	0.435**	0.001	0.025	−0.065	−3.395***
	(0.01)	(0.79)	(0.96)	(0.45)	(−0.46)	(0.24)	(0.45)	(−0.50)	(2.25)	(0.00)	(0.13)	(−0.29)	(−7.50)
Volatility	−0.300	−0.857**	−0.176	−0.192	−0.612**	−0.179	−0.387*	−0.185	−0.114	0.327	−0.500**	0.635***	1.393**
	(−0.79)	(−2.23)	(−0.62)	(−0.74)	(−2.32)	(−0.78)	(−1.73)	(−0.83)	(−0.48)	(1.53)	(−2.49)	(3.04)	(2.58)
Size	1.453***	0.920**	0.467	−0.258	0.293	0.310	0.693***	0.431*	0.133	0.071	1.187***	−0.072	−2.970***
	(3.52)	(2.50)	(1.47)	(−0.98)	(1.11)	(1.24)	(2.95)	(1.81)	(0.63)	(0.34)	(5.05)	(−0.27)	(−4.38)
BTM	0.218	0.304	0.026	0.135	−0.038	0.050	0.192	0.079	0.087	−0.340**	0.610***	−0.460***	−0.025
	(0.77)	(1.16)	(0.11)	(0.77)	(−0.21)	(0.33)	(1.26)	(0.50)	(0.58)	(−2.48)	(4.35)	(−3.17)	(−0.07)
Ret_{d-1}	−1.471***	−0.536*	0.093	−0.162	−0.594***	−0.046	0.012	−0.083	0.325**	0.259	0.439***	0.728***	−0.177
	(−4.61)	(−1.91)	(0.42)	(−0.85)	(−2.95)	(−0.27)	(0.08)	(−0.49)	(1.98)	(1.60)	(2.59)	(3.92)	(−0.44)
$Ret_{d-20,d-2}$	−0.323	−0.301	−0.022	−0.037	0.180	0.085	−0.026	−0.406**	−0.080	−0.249*	−0.101	0.584***	−0.618
	(−0.90)	(−1.09)	(−0.10)	(−0.18)	(0.96)	(0.48)	(−0.16)	(−2.43)	(−0.50)	(−1.69)	(−0.62)	(3.55)	(−1.35)
$Ret_{d-120,d-21}$	0.078	−0.322	0.110	0.007	0.105	0.147	−0.089	0.270	−0.067	−0.004	0.266	0.359*	2.212***
	(0.22)	(−0.91)	(0.37)	(0.03)	(0.45)	(0.77)	(−0.47)	(1.34)	(−0.33)	(−0.02)	(1.51)	(1.92)	(3.91)
# of Obs.	971,847	822,072	787,277	751,752	724,691	698,683	678,914	666,378	665,213	670,744	679,417	712,077	808,157
Adj. R^2	0.0979	0.1045	0.0964	0.0889	0.0915	0.0859	0.0855	0.0869	0.0838	0.0794	0.0808	0.0645	0.0934

Note. This table shows the Fama and MacBeth (1973) regression results of OOI predicting subsequent half-hour returns. Columns (1) to (12) show the results of the following equation for 12 half-hour intraday intervals: $Ret_{i,d,[t,t+1)} = \alpha + \beta \times OOI_{i,d,t} + \psi' X_{i,d,t} + \epsilon_{i,d,[t,t+1)}$, and column (13) shows the result for the following equation for the last half-hour interval: $Ret_{i,d,\mathrm{ON}} = \alpha + \beta \times OOI_{i,d,t} + \psi' X_{i,d,t} + \epsilon_{i,d,\mathrm{ON}}$, where $Ret_{i,d,[t,t+1)}$ denotes the returns of stock i on day d, estimated by the log difference of the NBBO quote midpoints at time t+1 and t, expressed in basis points. $Ret_{i,d,\mathrm{ON}}$ denotes the overnight return of stock i, estimated by the log difference between the first NBBO quote midpoints after 9:30 on day d+1 and the closing price on day d, expressed in basis points. $OOI_{i,d,t}$ is the options order imbalance using the trading volume of options underlying stock i on day d from time $t-1$ to t. $X_{i,d,t}$ denotes a battery of control variables: $SOI_{i,d,t}$ is the stock order imbalance using the trading volume of stock i on day d from time $t-1$ to t. $StockVol_{i,d,t}$ and $OptionVol_{i,d,t}$ are the natural trading volume of stocks i and options underlying stocks i on day d from time $t-1$ to t, respectively. $Spread_{i,d,t}$ denotes the average 1-minute quote spread of stock i on day d from time $t-1$ to t. $Turnover_{i,d}$ is the monthly average of daily turnover of stock i at the most recent month. $Volatility_{i,d}$ is the annualized volatility of stock i at the most recent month. $Size_{i,d}$ is the natural logarithm of market capitalization of stock i at the most recent quarter end. $BTM_{i,d}$ is the natural logarithm of equity book to market ratio of stock i at the most recent quarter end. $Ret_{i,d\text{-}1}$ ($Ret_{i,d\text{-}20,d\text{-}2}$ and $Ret_{i,d\text{-}120,d\text{-}21}$) is the daily (cumulative) returns of stock i on day d-1 (over day $d-20$ to $d-2$ and $d-120$ to $d-21$). All continuous independent variables are standardized for ease of interpretation. The robust t-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

Table OA4. Horse-Racing OOI in Predicting Next-Day Stock Returns

OOI	(1) t = 10:00	(2) t = 10:30	(3) t = 11:00	(4) t = 11:30	(5) t = 12:00	(6) t = 12:30	(7) t = 13:00	(8) t = 13:30	(9) t = 14:00	(10) t = 14:30	(11) t = 15:00	(12) t = 15:30	(13) t = 16:00
s = 10:00	1.211***	1.011**	0.775	0.513	0.164	0.361	−0.008	−0.119	0.593	0.962	1.332	1.835	2.035
	(4.00)	(2.48)	(1.58)	(0.90)	(0.26)	(0.51)	(−0.01)	(−0.13)	(0.62)	(0.93)	(1.20)	(1.53)	(1.53)
s = 10:30		1.163***	0.784	0.616	0.351	0.345	0.504	0.042	−0.694	−0.919	−0.977	−0.877	−1.191
		(2.85)	(1.64)	(1.14)	(0.57)	(0.49)	(0.68)	(0.05)	(−0.78)	(−0.94)	(−0.94)	(−0.77)	(−0.96)
s = 11:00			1.202***	0.992*	0.536	0.642	0.302	0.199	0.463	1.121	0.863	0.691	0.427
			(2.66)	(1.92)	(0.93)	(1.01)	(0.43)	(0.26)	(0.53)	(1.13)	(0.81)	(0.57)	(0.35)
s = 11:30				0.935*	0.341	0.213	0.755	0.938	0.910	0.553	0.638	0.353	−0.105
				(1.90)	(0.58)	(0.32)	(1.06)	(1.19)	(1.06)	(0.59)	(0.60)	(0.31)	(−0.09)
s = 12:00					1.403**	0.983	0.775	0.839	1.202	0.828	0.301	−0.107	0.534
					(2.54)	(1.61)	(1.16)	(1.07)	(1.43)	(0.90)	(0.30)	(−0.09)	(0.45)
s = 12:30						1.596**	1.501**	1.508*	1.022	1.087	1.002	0.486	0.691
						(2.43)	(2.06)	(1.86)	(1.18)	(1.19)	(1.01)	(0.46)	(0.61)
s = 13:00							0.431	−0.332	−0.767	−0.834	−1.577*	−1.677*	−1.542
							(0.63)	(−0.47)	(−0.97)	(−1.02)	(−1.74)	(−1.71)	(−1.47)
s = 13:30								1.773**	1.942**	2.084**	2.290**	2.835***	2.723**
								(2.39)	(2.29)	(2.33)	(2.48)	(2.88)	(2.54)
s = 14:00									0.745	0.711	0.109	−0.096	0.512
									(0.98)	(0.85)	(0.12)	(−0.10)	(0.46)
s = 14:30										0.068	−0.123	−0.257	−0.984
										(0.07)	(−0.12)	(−0.24)	(−0.86)
s = 15:00											2.351**	1.575	0.972
											(2.44)	(1.48)	(0.87)
s = 15:30												0.293	0.025
												(0.29)	(0.02)
s = 16:00													**3.911*** **
													(3.38)

Note. This table shows the Fama and MacBeth (1973) regression results of horse racing OOI over thirteen 30-minute intervals in predicting next-day stock returns:

$$Ret_{i,d+1} = \alpha + \sum_{s=10:00}^{t} \beta_s \times OOI_{i,d,s} + \psi' X_{i,d,t} + \varepsilon_{i,d+1}$$

In the first row, the t denotes the end time point of each regression, while the s in the first column shows the coefficients of each included OOI. All continuous independent variables are standardized for ease of interpretation. The robust t-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

Table OA5. Additional Analyses

Panel A: Open-buy put-call ratio			
Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
OBPC	−2.044***	−1.860***	−2.667***
	(−5.17)	(−5.33)	(−7.03)
Control	Yes	Yes	Yes
# of Obs.	780,624	702,251	691,802
Adj. R^2	0.1038	0.1047	0.1047
Panel B: Stock jump predictability			
Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
OOI	0.510***	0.377**	0.766***
	(3.83)	(2.51)	(5.21)
Control	Yes	Yes	Yes
# of Obs.	961,055	886,712	855,731
Adj. R^2	0.2359	0.2437	0.2444
Panel C: Stock crash risk predictability (realized skewness)			
Time interval =	*Opening*	*Midday*	*Closing*
	(1)	(2)	(3)
OOI	−0.044*	−0.000	0.052**
	(−1.95)	(−0.02)	(2.38)
Control	Yes	Yes	Yes
# of obs.	1,144,654	1,052,962	1,014,864
Adj. R^2	0.0630	0.0640	0.0648

Note. This table shows additional tests on informed options trading. In Panel A, the dependent variable is the next-day stock returns, and the main independent variable is options open-buy put-call ratio following Pan and Poteshman (2006). In Panel B, the dependent variable is stock jump size following Aït-Sahalia, Li, and Li (2025). In Panel C, the dependent variable is stock crash risks, measured by annualized realized skewness of 30-minute stock returns. All continuous independent variables are standardized for ease of interpretation. The robust *t*-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

Table OA6. Robustness Checks

	(1) t = 10:00	(2) t = 10:30	(3) t = 11:00	(4) t = 11:30	(5) t = 12:00	(6) t = 12:30	(7) t = 13:00	(8) t = 13:30	(9) t = 14:00	(10) t = 14:30	(11) t = 15:00	(12) t = 15:30	(13) t = 16:00
Panel A: Additional control for stock borrow fee													
OOI	1.310***	1.310***	1.054***	1.449***	1.987***	1.446***	1.703***	1.607***	1.103***	1.571***	1.899***	2.079***	3.449***
	(4.34)	(4.09)	(3.36)	(4.16)	(6.09)	(3.98)	(5.10)	(4.74)	(3.19)	(4.72)	(5.41)	(5.85)	(9.55)
BorrowFee	−4.887***	−5.145***	−5.071***	−5.847***	−5.249***	−6.140***	−5.633***	−5.239***	−5.918***	−6.020***	−6.178***	−6.127***	−5.793***
	(−7.34)	(−6.91)	(−6.86)	(−7.52)	(−6.86)	(−7.98)	(−7.06)	(−6.62)	(−7.03)	(−7.56)	(−7.74)	(−7.66)	(−7.63)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	962,772	814,085	779,564	744,357	717,375	691,529	671,955	659,586	658,486	663,917	672,541	704,991	800,321
Adj. R^2	0.1148	0.1214	0.1222	0.1246	0.1240	0.1255	0.1260	0.1280	0.1256	0.1251	0.1250	0.1225	0.1178
Panel B: Pooled Regression													
OOI	1.209***	1.338***	1.032***	1.458***	2.028***	1.607***	1.753***	1.816***	1.114***	1.574***	2.186***	2.198***	3.733***
	(3.50)	(3.68)	(2.93)	(3.88)	(5.58)	(4.24)	(4.46)	(4.78)	(3.01)	(4.08)	(5.66)	(5.96)	(9.34)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	970,001	820,532	785,860	750,394	723,396	697,373	677,650	665,159	664,043	669,622	678,197	710,882	806,731
Adj. R^2	0.2423	0.2438	0.2427	0.2416	0.2418	0.2413	0.2404	0.2404	0.2397	0.2378	0.2391	0.2364	0.2324
Panel C: Alternative Fixed Effects													
OOI	1.237***	1.383***	0.871**	1.182***	1.805***	1.331***	1.705***	1.737***	0.892**	1.591***	2.117***	2.077***	3.823***
	(3.45)	(3.64)	(2.35)	(2.95)	(4.77)	(3.30)	(4.02)	(4.31)	(2.19)	(3.88)	(5.16)	(5.30)	(9.23)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	807,100	672,416	641,212	609,865	585,531	562,602	545,438	534,323	533,365	537,914	545,803	574,381	660,438
Adj. R^2	0.2398	0.2411	0.2398	0.2388	0.2383	0.2384	0.2362	0.2364	0.2365	0.2339	0.2349	0.2324	0.2284

(cont'd)

Table OA6. Robustness Checks (cont'd)

Panel D: CBOE Data Only													
OOI_CBOE	1.407***	1.174***	1.042***	1.200***	1.697***	1.445***	1.532***	1.624***	1.056***	1.432***	1.859***	2.087***	3.266***
	(4.61)	(3.58)	(3.56)	(3.47)	(5.19)	(3.97)	(4.47)	(4.85)	(2.96)	(4.50)	(5.23)	(5.94)	(9.18)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	970,043	820,604	785,946	750,477	723,493	697,473	677,756	665,272	664,148	669,707	678,309	710,980	806,798
Adj. R^2	0.1098	0.1161	0.1167	0.1185	0.1183	0.1195	0.1201	0.1216	0.1196	0.1192	0.1186	0.1164	0.1123
Panel E: ISE Data Only													
OOI_ISE	0.108	1.072	0.318	0.740*	0.265	−0.083	0.186	0.706	0.351	1.562*	0.683**	0.837**	1.286***
	(0.35)	(1.53)	(0.85)	(1.95)	(0.37)	(−0.13)	(0.32)	(1.44)	(1.01)	(1.90)	(2.17)	(2.37)	(3.56)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Obs.	970,043	820,604	785,946	750,477	723,493	697,473	677,756	665,272	664,148	669,707	678,309	710,980	806,798
Adj. R^2	0.1095	0.1157	0.1164	0.1182	0.1181	0.1191	0.1199	0.1214	0.1191	0.1191	0.1182	0.1160	0.1117

Note. This table shows the robustness checks of OOI predicting subsequent returns. Panel A repeats the regression specifications in Table 2, additionally controlling stock borrow fees. The stock borrow fee (in %) is the indicative fee collected from Markit. Panel B repeats the regression specifications in Table 2 using pooled regression with firm and day fixed effects. Panel C repeats the regression specifications in Table 2 using pooled regression with industry and day fixed effects. Panels D and E repeat the Fama and MacBeth (1973) regression specifications in Table 2 using CBOE data and ISE data only, respectively. All continuous independent variables are standardized for ease of interpretation. In panel A, D, and E, the robust *t*-statistics based on Newey and West (1987) standard errors with 6 lags are shown in parentheses. In panel B and C, the robust *t*-statistics based on robust standard errors clustered at the stock and day level are shown in parentheses. ***, **, and * denote statistical significance at 1%, 5%, and 10% level, respectively.

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